COMPETITIVE INTELLIGENCE & FUNDING ANALYSIS REPORT

HCSP Competitive Intelligence & Expanded Funding Sources

Four verified SCWO competitor profiles — 374Water, General Atomics EMS, Aquarden Technologies, Revive Environmental — plus 35 screened non-dilutive funding channels.

Competitive positioning, the scale gap, customer segmentation, and a three-track funding strategy for a modular supercritical water oxidation system.

Prepared September 2026

Table of Contents

Executive Summary 3

1. 374Water Inc. (AirSCWO) 3

1.1 Financials and backing 4

1.2 Customers: real vs. pipeline 5

1.3 Scale limitations and target-customer attributes 6

2. General Atomics EMS (PERSES) 6

2.1 Backing: patient private capital inside a defense prime 7

2.2 Customers and deployments: >6M gallons destroyed, checkered record 7

2.3 Scale: 3 gpm floor, no small product, no commercial model 8

3. Aquarden Technologies (SuperOx) 9

3.1 Financials: family-owned, loss-making, equity-negative 9

3.2 Customers: two verifiable industrial systems, one controlling partner 10

3.3 Scale: 5 t/day ceiling, parallelization only, capacity the stated obstacle 11

4. Revive Environmental (Battelle PFAS Annihilator) 11

4.1 Backing: Battelle plus Viking Global, seed-stage revenue 11

4.2 Customers: four state takebacks, one lost state, two fixed sites 12

4.3 Scale: ~500 gal/day units and freight economics that exclude small generators 13

5. The Scale Gap and Customer Segmentation 13

5.1 Cross-company position 13

5.2 The evidence for the gap 14

5.3 What HCSP must do to hold the opening 14

6. Expanded Non-Dilutive Funding Channels 15

6.1 Directly actionable channels 16

6.2 Channels accessible via a host partner 17

6.3 Not accessible — ruled out, not padded 18

6.4 Sequencing with the existing 80-program register 19

References 20

Executive Summary

The four-company verdict: no credible supercritical water oxidation (SCWO) competitor currently serves the small, modular, on-site segment — and each is structurally prevented from doing so. The field splits into three failure modes. 374Water, the only publicly traded pure-play SCWO vendor, is financially fragile: FY2025 revenue of $215,037 against a $21.0M net loss, $1.78M cash at June 30, 2026, and explicit going-concern doubt in both its FY2025 10-K and Q2 2026 10-Q.[^W-2^][^W-3^][^W-6^] Aquarden Technologies, the only competitor with verifiable operating industrial SCWO plants, is technically insolvent on equity (−DKK 30.9M at end-2025), has lost money every year from 2021 through 2025, and employs 11 people.[^A-1^] General Atomics Electromagnetic Systems (GA-EMS) runs PERSES as a marginal business-development niche inside a ~$3.2B private defense prime: its 2025–2026 U.S. field activity is entirely government-funded demonstrations, with no permanent commercial installation at a paying U.S. customer publicly verifiable.[^G-13^][^G-27^] Revive Environmental operates a centralized destruction service from two permitted fixed sites in Ohio and Michigan; its ~500 gal/day units and concentrate-haulage economics reach small generators only through episodic, state-aggregated collection events.[^R-16^][^R-39^]

The scale floor is the tell. The smallest capital equipment deal evidenced in the field is 374Water's $4.88M Olathe project.[^W-6^] GA's smallest product is a 3 gpm industrial container with no published pricing or service model.[^G-9^] Aquarden offers a single ~5 t/day reactor class, scalable only by parallelization, in the $2–10M capex band.[^A-6^][^A-21^] Revive sells per-gallon destruction, not equipment, engineered for large or government-aggregated volumes.[^R-28^] No competitor offers a sub-$1M, small-facility, sold-and-self-operated on-site system. This is HCSP's opening — but it is not guaranteed to stay open. Four watch items could close it: a DIU Success Memo converting government demos into multi-site DoD orders (374Water already holds one; GA's is pending), ArianeGroup's ELIXIR service brand capturing Aquarden's channel, a Revive pivot to containerized on-site deployment, and Aquagga's HALT process undercutting SCWO on price at the state-program level.[^W-28^][^A-27^][^R-24^][^R-25^]

On funding, this report adds 35 screened channels beyond the existing US federal/state/municipal register. Eight are directly actionable for a US for-profit today — led by the Asian Development Bank Technology Innovation Challenge (up to $450K pilot grants), the AB InBev 100+ Accelerator (up to $100K pilots with five beverage majors), WaterEquity's Water & Climate Resilience Fund ($2–15M minority investments), and the GEF Challenge Program for Adaptation Innovation.[^N-15^][^N-34^][^N-26^][^N-4^] Nineteen more are accessible via a host partner; six are candidly ruled out. The practical implication: HCSP's non-dilutive strategy should now run on three tracks — US programs (existing register), multilateral/development-bank pilots for the islands and industrial-wastewater verticals, and corporate accelerator/procurement channels for the distillery and data-center verticals.

---

1. 374Water Inc. (AirSCWO)

Bottom line: 374Water is the closest thing HCSP has to a direct SCWO competitor — and it is one to two financing events away from distress. Its going-concern balance sheet, near-exhausted equity shelf, loss-making flagship contract, and one-at-a-time production mean it cannot service a long tail of small customers even if it

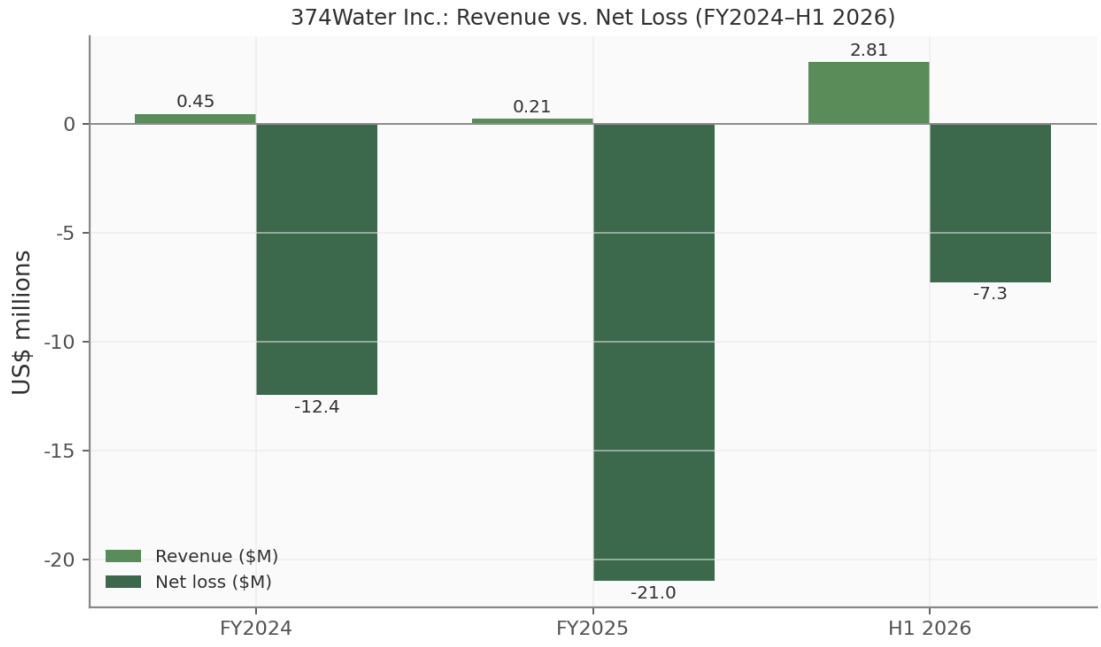
wanted to. HCSP should treat 374Water as validation of the technology class and as a counterparty-risk story to exploit in sales conversations, not as a pricing competitor.

1.1 Financials and backing

374Water is a Durham/Morrisville, North Carolina micro-cap (~$42–54M market capitalization, September–October 2026) commercializing SCWO under the AirSCWO brand, spun out of Duke University in 2018 and listed via a 2021 reverse merger.[^W-13^][^W-15^][^W-16^][^W-17^] The financial trajectory is one of declining revenue against rising losses, interrupted in 2026 by a single milestone event:

Metric	FY2024	FY2025	H1 2026
Revenue	$445,445 [^W-1^]	$215,037 [^W-2^]	$2,813,195 [^W-6^]
Net loss	$(12,434,114) [^W-2^]	$(20,975,052) [^W-2^]	$(7,267,858) [^W-6^]
Cash used in operations	$(10,589,735) [^W-2^]	$(14,326,205) [^W-2^]	$(2,297,376) [^W-6^]
Cash & equivalents (period end)	$10,651,644 [^W-2^]	$3,198,682 [^W-2^]	$1,776,259 [^W-6^]
Accumulated deficit	$28,387,618 [^W-1^]	$49,936,598 [^W-3^]	$57,204,456 [^W-6^]

All figures from SEC filings (FY2024/FY2025 Forms 10-K; Q2 2026 Form 10-Q). FY2025 revenue includes a negative adjustment of ~$1.9M from an accounting-estimate change on the OC San contract.[^W-2^]



374Water revenue vs. net loss, FY2024–H1 2026. Source: 374Water SEC filings (Forms 10-K and 10-Q).[^W-2^][^W-6^]

Three features matter more than any single number. First, **revenue quality is poor**: FY2025's $215,037 included a ~$1.9M negative adjustment tied to OC San delays, and H1 2026's apparent recovery rests on roughly $2.0M from a single event — passing the OC San Factory Acceptance Test; Q1 2026 revenue was only $551,155.[^W-

~$1.14M of stock actually sellable as of the August 2026 prospectus; the company has resorted to $860,000 of 10% convertible notes with warrants and a ~$630,000 secured note.[^W-3^][^W-6^][^W-7^] Third, **the largest shareholder is selling**: co-founder Yaacov Nagar (18.06%) executed 26 sales totaling 1,322,292 shares (~$1.29M) in six months, which the 10-K itself blames for depressing the share price; a 1-for-10 reverse split in December 2025 was required to retain the Nasdaq listing.[^W-3^][^W-9^][^W-12^][^W-18^] With $1.78M cash against a $2.3M half-year burn, runway is measured in months — exactly what the going-concern notes in both the FY2025 10-K and Q2 2026 10-Q state verbatim.[^W-3^][^W-6^] For HCSP the implication is direct: every competitive bid against 374Water should surface vendor-survivability risk, performance-bond incapacity, and milestone-funded manufacturing as procurement criteria.

1.2 Customers: real vs. pipeline

374Water's customer base is real but thin, and every dollar of it comes from government or government-adjacent buyers:

Customer	Deal	Value	Status (Sept 2026)
City of Orlando, FL (Iron Bridge WRF)	Demo ($812K, 2024) → 5-year Waste Destruction Services (WDS) license, March 2026	Demo $812K; ~$482K/qtr service revenue; company projects $3–5M/yr at initial scale	**Operating** — the flagship and only fixed-site WDS hub [^W-6^][^W-8^][^W-21^] [^W-23^]
Orange County Sanitation District (OC San), CA	AirSCWO 6 equipment sale, signed 2022	~$4.6M implied; district approved an $8.7M project budget in 2021	**~4 years late**; FAT passed June 2, 2026; delivery planned October 2026; **loss contract** — $1.6M accrued loss provision, $1.9M costs in excess of billings [^W-2^] [^W-5^][^W-6^][^W-20^] [^W-22^]
City of Olathe, KS (via Garney Construction)	AS6 + pretreatment/dewatering equipment	**$4,880,000 firm fixed price** ($3.0M unit); $2.2M milestone received Q1 2026	**Contracted, not yet built** — largest equipment order to date [^W-6^][^W-8^]
City of St. Cloud, MN	Mobile WDS pilot	$600,000	**Completed Sept 15, 2026**; grant-funded (Minnesota LCCMR/ENRTF) [^W-24^]
State of North Carolina / UNC Collaboratory	AFFF destruction	~$1M initial; up to 29,000 gal potential	Processing begun [^W-26^]

Announced pipeline that is **not** recurring revenue: a paid DoD/DIU demonstration at Clean Earth's Detroit facility (~900 gallons over 80 runtime hours, >99.9% destruction verified by Arcadis) that earned a DIU "Success Memo" in August 2026 creating a federal procurement pathway; an ESTCP demo at Peterson Space Force Base; a 2022 US Navy NDCEE award; a Crystal Clean siting agreement with no plant built; an Arcadis MOU; and a claimed "identified demand totaling nearly $1.8BN" that aggregates backlog, demos, and lab studies rather than booked orders.[^W-2^][^W-4^][^W-5^][^W-8^][^W-27^][^W-28^][^W-29^]

The pattern to act on is the gap between announced demand and recognized revenue. Five years into commercialization, 374Water has one operating hub (Orlando), one delivered-after-four-years equipment contract that loses money (OC San), one unbuilt equipment order (Olathe), and grant-funded pilots. The OC San episode — contract 2022, delivery October 2026, repeated timeline resets, a slipped Factory Acceptance Test — is the single most useful competitive artifact available to HCSP: it shows what happens when a thinly capitalized vendor attempts municipal-scale SCWO delivery, and it explains why buyers structure deals as evaluations with option-like expansions rather than commitments.[^W-12^][^W-21^][^W-22^]

1.3 Scale limitations and target-customer attributes

Only two AirSCWO hardware classes exist: the AS1 (1 wet ton/day, mobile, first unit completed 2025–26) and the AS6 (6–10 tons/day, the only commercially operating class). The AS30 is under development with first manufacturing "late 2026"; the AS100+/AS200 remains a slide-level concept described as "bespoke, to be developed in the future based on customer orders." The 10-K itself narrows the marketing claim of "1 to over 200 tons per day" to an operating line "designed to scale between six (6) and ten (10) tons per day."[^W-3^][^W-4^][^W-30^]

Three structural facts define the scale floor. **First, the smallest capital deal in the field is ~$4.9M** — the Olathe AS6 at $3.0M for the unit plus pretreatment, fees, and warranty.[^W-6^] **Second, the nominally small AS1 is not sold as a product**; it is kept in 374Water's own WDS fleet for crew-operated $0.5–1.5M/yr municipal pilots, marketed against 0.25–0.5 MGD plants (towns of ~2,500–5,000+ people).[^W-4^][^W-24^] **Third, economics force scale-up**: management's deck pegs WDS site capex at $1.5–10M with payback pitched only at hub scale, and the September 2026 strategy announcement explicitly targets regional centralized facilities aggregating biosolids from multiple plants — a wholesale, utility-scale model.[^W-4^][^W-32^] Founder Nagar's own framing confirms it: the smallest containerized unit was "sized for a community of about 6,000 people," reaching net energy production only at ~30,000-person scale.[^W-31^]

The target-customer profile follows mechanically: large municipal utilities (OC San at 180 MGD, Orlando at 40 MGD, Olathe at 11 MGD), federal defense programs, and hazardous-waste treatment, storage, and disposal facilities as channel hosts — all multi-year, grant-anchored, government-funded procurements.[^W-5^][^W-12^][^W-24^][^W-28^] The segments 374Water does not serve are precisely HCSP's beachhead: small industrial generators, farms, food and beverage plants, remote sites, and any private buyer without grant funding. No named private industrial customer exists anywhere in the filings or press record, and the original Gates Foundation "Global Access" small-sanitation mission has been abandoned in practice.[^W-2^][^W-4^][^W-14^] With 48 employees, one factory, and two systems in existence at year-end 2025, the constraint is not marketing choice — it is capital structure and bandwidth.[^W-3^][^W-17^]

---

2. General Atomics EMS (PERSES)

Bottom line: GA is the field's technology incumbent and its commercial ghost. Thirty-six years of development, ~$250M cumulative investment, EPA-verified >99.99% PFAS destruction, and >6 million gallons destroyed make PERSES the default "safe" choice for large government programs. But the line is a business-development-led niche inside a ~$3.2B private defense prime, its 2025–2026 U.S. activity is 100% government-funded

demonstrations, and its own anchor customer type has twice rejected its SCWO systems on capacity, reliability, and cost. GA is a competitor for national programs only — and a reputational cautionary tale HCSP can use.

2.1 Backing: patient private capital inside a defense prime

General Atomics has been owned by the Blue family since 1986, generates ~$3.2B in annual revenue (2024, Forbes/FactSet), holds $41.5B in cumulative federal contract obligations including a $14.1B MQ-9 IDIQ awarded September 2025, and employs ~15,000 people.[^G-1^][^G-3^][^G-4^] The SCWO line sits inside GA-EMS, a division whose strategic identity is electromagnetic aircraft launch systems, space, hypersonics, and directed energy; hazardous-waste remediation is a stated but marginal commercial-products tail.[^G-8^][^G-14^] Funding has been mixed internal R&D plus DoD program money over ~36 years — GA's own 2016 presentation cites "22 years and $250M of SCWO development… over 15 working SCWO systems… over 35,000 hours of steady state operations."[^G-9^][^G-10^] PERSES revenue is not broken out and is de minimis against enterprise revenue (single-digit $M per year at most, inferred from documented unit sales and demo contracts).

The strategic read: PERSES will not die for lack of capital — the Blue family's self-funding culture is famous — but one named business-development director carries the entire public-facing effort, the line was rebranded only in 2025 (trademark registered January 2026), and marketing runs through defense conferences rather than waste-industry channels.[^G-10^][^G-16^][^G-32^] The risk to customers is not vendor bankruptcy — the opposite of the 374Water and Aquarden situations — but deprioritization, slow delivery, and program dependence. HCSP cannot out-balance-sheet GA; it can out-iterate, out-price, and out-service it in segments GA's organization will never staff.

2.2 Customers and deployments: >6M gallons destroyed, checkered record

GA's verified claims are genuine: >6 million gallons of waste destroyed at >99.99% efficiency across 200+ waste streams, and the first-ever EPA test/verification of a commercial SCWO system destroying PFAS in aqueous film-forming foam (AFFF), documented in EPA report EPA/600/R-22/257 (September 2022).[^G-15^][^G-16^][^G-17^] The deployment record behind those claims is mixed:

Site / Program	Outcome	Evidence
Tooele Army Depot, UT — iSCWO for demil secondary waste	**Abandoned**	EPA (2019): "eventually was not used due to lack of capacity and reliability issues, and offsite treatment methods were found to be more cost effective" [^G-19^]
Blue Grass CAPP, KY — four-reactor SCWO for agent hydrolysate	**Built, never operated; rejected August 2020**	Hydrolysate shipped to Veolia instead; National Research Council had flagged salt-plugging, corrosion, and long-duration reliability concerns since 1999–2002 [^G-11^][^G-20^][^G-21^]
Camp Minden, LA — 3 gpm iSCWO for M6/propellant	**Collapsed**	Customer Explo Systems bankruptcy; unit stranded; 10 gpm follow-on redirected [^G-9^]

Kyrgyzstan & Tajikistan — transportable iSCWO, GEF-funded	Operating/recent	DDT destruction in 5,000 tonnes of waste under UNEP/Basel guidelines [^G-23^]
Tianjin, China — three iSCWO systems sold (2018)	Unknown follow-through	No public evidence of commissioning or follow-on orders [^G-24^]
First commercial sale 2012–2014 — customer unnamed	"First two systems are still in use" (GA, Oct 2025)	Identity not publicly disclosed [^G-10^]

Current activity is demonstrations, not operations. The DIU/ESTCP demo at Clean Earth Charlotte, NC ran six PFAS waste streams in August–September 2025; a USACE/ERDC-funded demo via Bay West ran three streams at Lake Elmo, MN in May–June 2025 before the unit rotated elsewhere. As of September 2026, **no public results report or DIU Success Memo for GA has been located** — a notable asymmetry, since 374Water received its Success Memo in August 2026.[^G-13^][^G-25^][^G-27^][^G-28^][^W-28^] No publicly verifiable permanent commercial PERSES installation exists at a paying U.S. customer's facility. One clarification for the record: the full-scale SCWO unit operating in France for the defense industry belongs to **Aquarden**, not GA — searches for any GA–ArianeGroup link returned nothing.[^G-33^]

The "so what" is two-sided. GA's EPA verification and pedigree dominate procurement narratives for large programs and make it the incumbent to beat in DoD/DOE channels. But Tooele and Blue Grass are government-published evidence that GA's SCWO was rejected by its own anchor customer type on capacity, reliability, and cost grounds — the "graveyard pattern" that haunts the technology class and that HCSP's reliability-documentation strategy (Section 5.3) exists to escape.

2.3 Scale: 3 gpm floor, no small product, no commercial model

GA's baseline module is a 3 gpm (~4,300 gal/day hydraulic) containerized skid — 25 ft × 8 ft × 9.5 ft, ~20,800 lb — scalable only by parallel modules, operating at 650 °C / 4,000 psi and requiring a tail of air compression, NaOH quench, and pre/post-processing.[^G-9^][^G-10^] Larger fixed units exist historically (a 10 gpm unit; the 2,700 kg/h four-reactor Blue Grass design), but there is **no sub-3 gpm product, no published pricing, no per-gallon service model, and no financed or leased offering**.[^G-9^][^G-22^] Realized deployments require RCRA-permitted host sites or government sponsorship; the Basel Convention guidelines themselves note SCWO units "are predominantly used in fixed configurations."[^G-13^][^G-23^] GA positions PERSES explicitly against "lab and prototype systems" — competing on industrial scale and maturity, which is a declaration that the small end is not its market.[^G-16^]

Target customers confirm the shape: DoD environmental programs (ESTCP, DIU), USACE/ERDC via environmental primes, large TSDFs as demonstration hosts, international aid-funded cleanups, and occasional industrial sales abroad.[^G-13^][^G-23^][^G-24^][^G-27^] Absent entirely: municipal utilities buying directly, small industrial generators, landfill operators as equipment buyers, and any throughput-priced commercial service. The verdict: GA is a real competitor for large government and national-program business only, and its corporate structure makes it unlikely to contest the small/modular segment absent an external trigger such as a major DoD production contract funding a smaller standardized product. That trigger — a DIU Success Memo for the Charlotte demo — is the most important watch item in this profile.[^G-13^]

---

3. Aquarden Technologies (SuperOx)

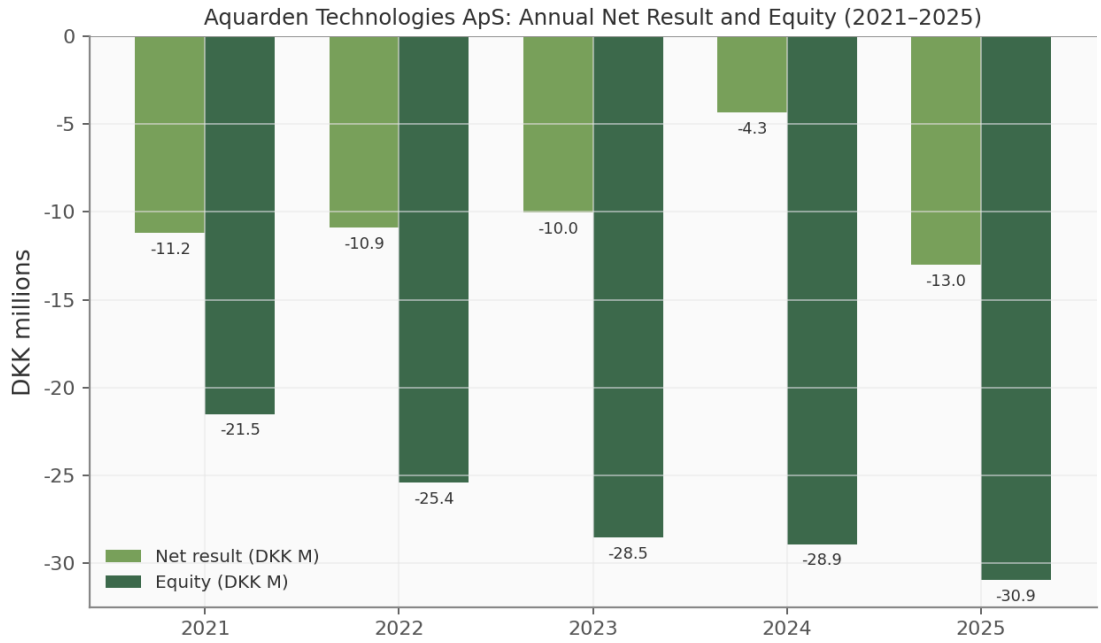
Bottom line: Aquarden owns the scarcest asset in the field — verifiable operating industrial SCWO plants — and the weakest balance sheet. Negative equity of −DKK 30.9M, five consecutive loss years totaling ~DKK 49.5M, 11 employees, and total dependence on one customer relationship (ArianeGroup) make it technically insolvent on equity and commercially captive. The Danish EPA's own conclusion that "the main obstacle now for onsite implementation is the capacity and throughput" is the competitor's scale ceiling stated by its own regulator.

3.1 Financials: family-owned, loss-making, equity-negative

Aquarden Technologies ApS (CVR 29151911, Skævinge, Denmark; founded 2005) is family-owned by founder and sole director Zhuoyan Cai through Cai Holding; no institutional or venture investors appear in registry data. [^A-2^][^A-3^] Danish registry financials (in DKK '000) show a company shrinking into its commercial phase:

DKK '000	2021	2022	2023	2024	2025
Gross profit	1,795	−532	1,111	5,843	**−1,528**
Net result	−11,177	−10,886	−10,029	−4,343	**−13,025**
Equity	−21,505	−25,391	−28,520	−28,913	**−30,947**
Cash	4,639	4,467	1,198	1,213	2,706
Employees (avg.)	13	13	8	10	**11**

Source: CVR/Virk annual reports via Proff.dk. Net revenue is not disclosed under the Danish small-company exemption; gross profit of DKK 1.1–5.8M in the best years implies revenue realistically in the low tens of millions DKK (≈ €1–4M) at most — estimate, flagged.[^A-1^]



Aquarden net result and equity, 2021–2025. Source: Danish CVR registry annual reports via Proff.dk.[^A-1^]

The trajectory worsened in 2025: the largest loss on record (−DKK 13.0M), a negative gross profit (direct project costs exceeded billings), and cash of DKK 2.7M against roughly DKK 7.3M of annual payroll — months of runway absent new owner injections.[^A-1^] Operations are carried by ~DKK 77M of growing “other debt” plus ~DKK 6.8M of related-party debt — owner-loan financing, not external equity.[^A-1^] Product maturation has been financed by grants: the Danish EPA’s MUDP 2020 “PFAS-inator” project and €1.59M of French state aid on the €5.03M ArianeGroup ELIXIR project.[^A-10^][^A-15^] Danish trade press characterized the company as a “deficit-hit cleantech company” as early as 2021, and under Danish law persistent negative equity requires owner support to maintain going concern.[^A-17^] The practical implication for HCSP: Aquarden cannot finance customer-facing commercial models (leasing, build-own-operate), cannot build an international service network, and is one disrupted relationship away from crisis — which makes its technology credibility available to acquirers and its customers available to better-capitalized rivals.

3.2 Customers: two verifiable industrial systems, one controlling partner

Aquarden’s operating commercial fleet consists of exactly two industrial SuperOx units, both at ArianeGroup sites in France — the first (2018) destroying energetic wastes, the second (2024) for PFAS destruction — plus Aquarden’s own Danish full-scale plant used for customer test campaigns.[^A-5^][^A-8^] Everything else in the public record is pilots and treatability tests: Stockholm Arlanda Airport AFFF concentrate (2020), Sörab landfill leachate (2019–20), a US EPA-directed AFFF demonstration run on Aquarden’s **lab-scale** system in Denmark at 1 L/h (2021) — in which Aquarden’s effluent PFOS residual of 240 µg/L was the highest of the three vendors tested — and a Royal Copenhagen Hospital pharmaceutical-wastewater demonstration.[^A-7^][^A-11^][^A-12^][^A-13^] The company’s claim of deployment “in more than five countries in Europe” conflates test campaigns with installations.[^A-19^]

delegated to small partners (PFAS Busters LLC in Japan, treatment-train partners such as ECT2/Vandrensning.com in Denmark).[^A-7^][^A-9^][^A-15^][^A-19^][^A-27^] Aquarden has **no US presence**: the EPA-directed test ran in Denmark, the CEO's 2023 statement that the company is "in talks with U.S. companies" remains aspirational, and it is absent from every US DoD/ESTCP program.[^A-11^][^A-18^] For HCSP, Aquarden is therefore not a US-market competitor at all; it is a European technology supplier whose customer-facing brand is being absorbed by a defense prime — a live demonstration of what happens to an undercapitalized SCWO vendor that anchors on a single channel partner.

3.3 Scale: 5 t/day ceiling, parallelization only, capacity the stated obstacle

The SuperOx reactor processes ~5 tonnes/day (up to 12 m³/day of liquid waste or slurry, ~180 L/h continuous) at 80–300 kWh/m³, all-electric; scale-up is only via parallelization of identical reactors — no larger reactor size exists.[^A-6^][^A-7^][^A-15^] Third-party assessment puts SCWO units of this class at ~$2–10M capex per unit with destruction costs of ~$100–500/tonne of concentrate.[^A-21^] The decisive evidence is the Danish EPA's PFAS-inator final report: SCWO works technically, but *"the main obstacle now for onsite implementation is the capacity and throughput"* — and it is therefore positioned as the expensive final step of a treatment train after concentration, not a standalone plant.[^A-15^][^A-16^] An Environment Analyst corporate guide reaches the same conclusion: SCWO "require[s] high capital investment and skilled operation, often limiting their early adoption to larger regional facilities or remediation contractors."[^A-20^]

Aquarden's target customers are large European defense/aerospace groups (ArianeGroup, essentially exclusively), plus airports, fire services, military sites, and industrial facilities with small-volume, high-concentration, high-liability streams — AFFF concentrate, foamate, spent sorbents, explosives.[^A-7^][^A-8^][^A-29^] Municipalities and small sites are reached only indirectly, as generators shipping concentrate to a hub or through partner treatment trains; there is no named municipal SuperOx purchase and no small-format product.[^A-15^] The honest qualification: Aquarden's destruction performance (>99.99% on PFAS concentrates and explosives at industrial scale in France) is genuinely demonstrated and rare among SCWO vendors — its moat is engineering credibility plus the ArianeGroup channel, not scale, cost, or reach.[^A-5^][^A-15^] HCSP's counter is economic, not technical: serve the streams and site sizes Aquarden's 5 t/day, $2–10M unit class structurally cannot.

---

4. Revive Environmental (Battelle PFAS Annihilator)

Bottom line: Revive is the best-backed and best-permitted SCWO player in North America — and it has chosen a business model that structurally skips small generators. Its centralized, per-gallon destruction service from two permitted fixed sites, fed by concentrate haulage and state-aggregated collection events, leaves on-site and small-facility demand open by design. Revive's "Certificate of Annihilation" has simultaneously become the procurement documentation bar HCSP must match.

4.1 Backing: Battelle plus Viking Global, seed-stage revenue

Revive Environmental Technology, LLC spun out of Battelle Memorial Institute in December 2022 (launched publicly January 17, 2023), jointly owned by Battelle and hedge fund Viking Global Investors "with each sharing governance"; no financial terms, valuation, or subsequent external rounds have been disclosed.[^R-1^] Battelle contributed the patented PFAS Annihilator (SCWO) and GAC Renew technologies after a decade of internal R&D

and retains strategic control — Battelle EVP Matt Vaughan chairs Revive's board.[^R-2^][^R-6^] BlueTech Research classifies the company as private, seed-stage, early revenue, with a service-based treatment-fee model. [^R-5^] Revenue is undisclosed; identifiable anchors — $668,258 (New Hampshire), $3M (Ohio Round 1), a $16.6M appropriation behind the New Jersey program — imply a low-to-mid tens-of-millions annual revenue services business at most (estimate, flagged).[^R-9^][^R-10^][^R-11^] Headcount is likewise undisclosed; a March 2025 plan to add 10–20 staff, framed as significant growth, implies a base of roughly 30–70 employees (estimate).[^R-8^]

The cautionary signals are specific. Founding CEO David Trueba (ex-Evoqua) stepped down March 6, 2026 — replaced by Chief Commercial Officer Rick Gillespie — a leadership transition three years post-launch, mid-scale-up.[^R-6^] Revenue concentration is episodic and grant-dependent: Ohio's Round 1 was funded by a one-time $3M Monsanto PCB settlement rather than recurring appropriations, and the Ohio EPA is publicly "trying to find funds" to destroy remaining stockpiles.[^R-10^][^R-19^] Growth requires building Annihilator units (five observed at Columbus), winning state-by-state facility permits — the Columbus site took until 2026 to become Ohio's first permitted PFAS-destruction facility — and funding concentration infrastructure ahead of revenue, all carried by the two owners, with no new external capital disclosed since launch.[^R-14^][^R-18^]

4.2 Customers: four state takebacks, one lost state, two fixed sites

Revive's visible revenue is state AFFF takeback programs: New Hampshire (first-in-nation, July 2023; 10,000 gal for $668,258, ~$67/gal all-in with haulage to Michigan), Ohio (Round 1 completed August 2026: 13,500+ gal from 120 fire departments and airports), North Carolina (statewide contract won March 2025 in a competitive pilot against AxNano, Clean Earth, and 374Water; Phase 1 completed January 2026 with third-party verified destruction), and New Jersey (NJDEP contract announced March 2026; ~150,000 gal from 400+ departments backed by a $16.6M appropriation).[^R-9^][^R-10^][^R-11^][^R-13^][^R-22^][^R-23^] **Colorado chose competitor Aquagga's HALT process at a $40/gal program payment** — evidence that SCWO is contestable on price at the state-program level.[^R-24^] Beyond takebacks: the 4never™ landfill-leachate consortium (Heritage-Crystal Clean, Allonnia, EPOC Enviro, Revive) destroying 300–500 gal/day of concentrate at a reported $0.15–0.40/gal of raw leachate; DoD ESTCP/DIU demonstrations including an April 2025 field demo at Clean Earth Detroit; and airport foam-transition work marketing a third-party "Certificate of Annihilation" verified to EPA Method 1633.[^R-16^][^R-25^][^R-28^][^R-29^][^R-35^] DoD positioning is prospective: February 2026 DoD guidance lists SCWO among "emerging options… [to] monitor" while naming landfills and incineration as established — Revive is not yet a default DoD disposal pathway despite the October 1, 2026 AFFF phase-out deadline creating a >2M-gallon disposal wave.[^R-26^][^R-27^]

Operations are centralized at two permitted fixed facilities — Columbus, OH (five Annihilator units as of EPA Administrator Zeldin's July 7, 2026 tour) and Grand Rapids, MI (co-located inside a Heritage-Crystal Clean site, operating since May 2023 as the first permitted commercial SCWO PFAS destruction in the US).[^R-14^][^R-16^][^R-17^] Although the units are containerized and were historically described as mobile, commercial destruction happens under facility permits: "We concentrate the PFAS on-site, then destroy the concentrate at our permitted facility."[^R-34^] The 2025 WEFTEC Innovative Technology Award citation — "North America's first fully permitted SCWO-based PFAS destruction system operating at commercial scale" — is both Revive's moat and its self-description as a facilities business, not an equipment vendor.[^R-20^]

4.3 Scale: ~500 gal/day units and freight economics that exclude small generators

Each Annihilator unit processes roughly 500 gal/day of concentrate (300–500 gpd at Grand Rapids); the company claims up to 7M gal/yr aggregate capacity.[^R-16^][^R-37^][^R-38^] The service model's physics exclude dispersed small generators from direct service: waste must be bulk-shipped as concentrate or concentrated on-site and hauled to Ohio or Michigan, with freight, chain-of-custody, and per-project overhead imposing a de facto minimum economic volume. Small generators are reached only through government aggregation — Ohio's collection events restrict drops to under 1,000 gal per waste owner.[^R-34^][^R-39^] Revive's own framing concedes the orientation: treatment is "cost-effective at scale for states, fire departments, airports, and commercial customers."[^R-14^] Independent evidence confirms the pattern industry-wide: DoD demo participants concluded on-site treatment beats off-site hauling on cost and logistics; GAC reactivation incumbents refuse site-specific contracts below ~36,000 kg per changeout; and the Minnesota PCA found small wastewater facilities face per-pound PFAS costs over six times those of large facilities.[^R-36^][^R-40^][^R-43^]

Revive's target customers are therefore state agencies as prime-contract counterparties, landfill operators via consortium, airports, DoD (prospectively), and claimed Fortune 500 industrials — procurement by service contract and per-gallon fee, not equipment sale.[^R-28^][^R-31^] The segments not served map directly onto HCSP's opening: small and remote generators whose volumes cannot justify Midwest freight; continuous dilute-stream producers at small scale (95% of US public water systems serve under 10,000 customers and lack staff and budget for treatment-plus-hauling strategies); customers wanting equipment ownership rather than chain-of-custody transport; and solids/biosolids generators, since Revive's SCWO handles liquids, slurries, and concentrates only.[^R-37^][^R-40^][^R-44^] One capability transfer HCSP must treat as mandatory: Revive's third-party-verified Certificate of Annihilation has become the de facto documentation requirement in state RFPs — North Carolina's program explicitly required verified destruction — and any HCSP offering without an equivalent certificate will be disqualified before price is discussed.[^R-13^][^R-29^]

---

5. The Scale Gap and Customer Segmentation

5.1 Cross-company position

Dimension	374Water (AirSCWO)	General Atomics EMS (PERSES)	Aquarden (SuperOx)	Revive (PFAS Annihilator)
Financial health	Going-concern doubt (FY25 10-K, Q2'26 10-Q); $1.78M cash; $57.2M deficit [^W-3^][^W-6^]	Parent ~$3.2B revenue, Blue family; line is a marginal BD niche [^G-3^][^G-4^]	Negative equity −DKK 30.9M; losses every year 2021–25; 11 staff [^A-1^]	Battelle + Viking Global; undisclosed, early-revenue, CEO replaced Mar 2026 [^R-1^][^R-5^][^R-6^]
Smallest offering	AS1, 1 t/d mobile — service-only, not sold [^W-4^]	3 gpm (~4,300 gal/d) container + balance-of-plant [^G-9^]	5 t/d reactor; parallelization only [^A-6^]	~500 gal/d concentrate unit, centralized [^R-16^]
Smallest deal evidenced	$4.88M equipment project (Olathe);			

	$600K crew-operated pilots [^W-6^][^W-24^]	None priced; demos government-funded [^G-13^][^G-27^]	$2–10M/unit capex class (third-party est.) [^A-21^]	Per-gallon service; small volumes only via state events (<1,000 gal caps) [^R-39^]
Customer types	Municipal utilities, state programs, DoD [^W-5^][^W-28^]	DoD/USACE programs, TSDF hosts, aid-funded cleanups [^G-13^][^G-23^]	European defense prime (ArianeGroup), partner treatment trains [^A-8^] [^A-15^]	State agencies, landfill consortium, airports, DoD demos [^R-9^][^R-28^]
Segments NOT served	Sub-$1M buyers, private industrials, small/rural sites [^W-4^][^W-32^]	Anything sub-3 gpm; all commercial service models [^G-9^][^G-16^]	Small facilities, bulk streams, US market entirely [^A-15^][^A-18^]	Small/remote/continuous generators; equipment buyers; biosolids [^R-37^][^R-44^]

The table's consistency is the finding. Four competitors with four different ownership structures — public micro-cap, private defense prime, family-owned SME, and nonprofit/hedge-fund joint venture — have independently converged on the same excluded customer: the small facility that wants destruction on its own site at a capital cost below $1M. That convergence is not a marketing coincidence; it reflects shared economics of the incumbent technology class — high-pressure reactor capex, skilled operation, permitting, and equipment-amortization dominance in industry cost projections of $12–90/ton — compounded by each firm's specific constraints.[^G-31^] A fifth column would show the same shape for adjacent technologies: Aquagga's HALT won Colorado at $40/gal as a centralized program service, not a small on-site product.[^R-24^] The gap is a market-structure fact, not a gap in any one company's lineup.

5.2 The evidence for the gap

Four independent scale floors bracket the underserved space. **Capital floor:** the smallest equipment transaction evidenced in the field is 374Water's $4.88M Olathe project; its smallest unit is not sold at all.[^W-4^][^W-6^] **Throughput floor:** GA's 3 gpm baseline (~4,300 gal/day hydraulic) is a full industrial plant requiring RCRA-permitted hosting or government sponsorship; no sub-3 gpm variant exists.[^G-9^][^G-13^] **Unit-class floor:** Aquarden's 5 t/day reactor in the $2–10M capex band is the minimum economic unit, and its own regulator names capacity and throughput as the obstacle to on-site implementation.[^A-15^][^A-21^] **Logistics floor:** Revive's model prices small generators out via freight and chain-of-custody unless a state aggregates them, and even then caps individual drops below 1,000 gallons.[^R-34^][^R-39^] The demand side is documented by regulators and the competitors' own ecosystems: 95% of US public water systems serve fewer than 10,000 customers; small facilities face per-pound PFAS costs over six times those of large ones; and DoD demonstration participants themselves concluded on-site treatment beats off-site disposal "because of the logistical and permitting challenges of moving waste from one place to another."[^R-40^][^R-43^][^R-44^] No credible sub-$1M, small-facility, on-site SCWO offering exists. That is the segment HCSP's modular architecture — self-fueling hydrogen loop, distilled-water and power/cooling co-products — is built for, and the co-product economics are precisely what can make sub-scale sites pay where competitors' disposal-cost math cannot.

5.3 What HCSP must do to hold the opening

The opening is real but time-limited; holding it requires five concrete moves, each mapped to a documented competitor failure or capability.

1. **Ship small and customized variants.** Every competitor's floor is set by a single industrial unit class; HCSP's product line should be explicitly sized below the $4.9M capital floor and below 3 gpm, with customization as a stated capability — the segment definition itself.[^W-6^][^G-9^]
2. **Price on-site destruction per gallon as a service option.** Revive's $0.15–0.40/gal (raw leachate) and ~$67/gal (AFFF all-in) benchmarks define the price umbrella; an on-site HCSP service that eliminates freight and haulage attacks the cost element — logistics — that DoD demo participants identified as decisive.[^R-9^][^R-35^][^R-40^]
3. **Issue third-party verified destruction certificates from day one.** Revive's Certificate of Annihilation (EPA Method 1633 verification) is now the de facto procurement bar in state RFPs; HCSP must match it or be excluded pre-price.[^R-13^][^R-29^]
4. **Document reliability relentlessly to escape the graveyard pattern.** GA's Tooele abandonment and Blue Grass rejection, plus 374Water's four-year OC San delivery and loss-contract provision, have primed buyers to expect SCWO delivery failure; published runtime hours, availability statistics, and reference-site uptime are competitive weapons, not marketing garnish.[^G-19^][^G-20^][^W-2^]
5. **Track four watch items with named owners:** (a) a **DIU Success Memo for GA's Charlotte demo** — 374Water already holds one (August 2026), and a GA memo would unlock sole-source-style DoD scaling; (b) **ArianeGroup ELIXIR expansion** (Toulouse planned) — channel capture of Aquarden's technology could produce a well-funded European service competitor; (c) a **Revive containerized pivot** — its units deployed on-site at the Detroit DoD demo, so a capitalized on-site service pivot is technically available; (d) **Aquagga HALT price competition** — Colorado's $40/gal award shows state programs will choose non-SCWO destruction on price.[^W-28^][^G-13^][^A-27^][^R-24^][^R-25^]

The common thread: HCSP's window is defined by competitors' structural constraints (capital, organization, permits, freight), not by their ignorance of the segment. Windows defined by constraints close when constraints lift — a 374Water recapitalization, a GA DoD production contract, a Revive on-site pivot — so speed to reference installations matters more than breadth of pipeline.

---

6. Expanded Non-Dilutive Funding Channels

Bottom line: beyond the existing 80-program US register, 35 additional channels were screened across multilaterals, UN/NGO structures, corporate and family foundations, and climate philanthropies. Eight are directly actionable for HCSP as a US for-profit today; nineteen require a host partner; six are ruled out. The highest-expected-value additions are the ADB Technology Innovation Challenge (the cleanest true grant available to a for-profit SCWO vendor), the AB InBev 100+ Accelerator (pilot cash plus entrée to five beverage majors), WaterEquity's Water & Climate Resilience Fund ($2–15M minority tickets), and the Water-AI Nexus at the University of Pennsylvania (the strongest non-dilutive entry into the data-center water vertical).

6.1 Directly actionable channels

#	Channel	Type	Size	Eligibility for HCSP	Cycle	Best-fit vertical
1	ADB Technology Innovation Challenge [^N-15^]	Pilot grant	Up to $450K (10% co-financing)	**Direct** — open to technology companies piloting in ADB developing member countries	Themed calls; monitor challenges.adb.org	Islands/Pacific; industrial wastewater — **top pick**
2	AB InBev 100+ Accelerator (with Coca-Cola, Colgate-Palmolive, Danone, Mondelēz, Unilever) [^N-34^][^N-35^]	Pilot funding + accelerator	Up to $100K/pilot, 6-month program	**Direct** — built for for-profit startups; water stewardship is a standing challenge	Annual cohort (~spring applications)	Distillery/brewery wastewater + water reuse
3	WaterEquity Water & Climate Resilience Fund [^N-26^][^N-27^]	Equity/mezzanine, minority stakes	$2–15M tickets; >$100M raised of $200M target (Microsoft anchor)	**Direct** — open application for water infrastructure and growth companies in 8 emerging markets	Rolling	C&I wastewater/reuse project SPVs
4	GEF Challenge Program for Adaptation Innovation [^N-4^][^N-5^]	Competitive grant	~$10M per call; >$40M committed across 32 projects	**Direct application** — the one GEF window inviting private-sector innovators; projects must deliver adaptation benefits in developing countries	Periodic calls (funds flowing through 2026)	Islands/remote climate-resilient water

5	Water Center at Penn — Water-AI Nexus (with AWS, WEF, Leading Utilities of the World) [^N-60^]	Research consortium / pilot cost-share	Project-negotiated	**Direct partnership** — industry collaborators join and co-fund pilots	Rolling	Data-center cooling/blowdown — **top pick for DC vertical**
6	WEF/WRF LIFT Technology Scans [^N-61^][^N-64^]	Utility consortium vetting (non-cash)	In-kind adoption risk-sharing; ~300 utilities	**Direct** — designed for technology vendors	Rolling/annual scans	Biosolids, PFAS, industrial wastewater
7	Breakthrough Energy Discovery — Explorer grants [^N-54^][^N-55^]	Philanthropic grant	$50–250K (TRL 0–2); Fellows track up to $500K as SAFE (dilutive)	**Direct but fit-constrained** — hard screen of ≥500M tonnes CO_2e/yr at scale; viable only via biosolids methane-avoidance + fossil-displacing power framing	Annual; next concept-paper deadline reported Nov 2, 2026	Biosolids/energy nexus only
8	ClimateWorks targeted RFPs [^N-52^][^N-53^]	Grant (RFP-specific)	Pool ~$9M (2025 Adaptation & Resilience Fund)	**Possible on specific RFPs** — eligibility has explicitly included private enterprises; no standing application	Episodic; monitor newsletter	Water resilience/adaptation

This table should be read as a portfolio, not a menu. The ADB and GEF adaptation windows monetize the islands/remote vertical, where HCSP's distilled-water and power co-products constitute an adaptation narrative pure-disposal competitors cannot replicate; both require developing-country deployment, so they pull HCSP's international pilot roadmap forward. The 100+ Accelerator and Water-AI Nexus are vertical-entry instruments: the pilot funding matters less than procurement entrée to five beverage majors and a consortium seat alongside AWS and the Water Environment Federation. WaterEquity is patient capital rather than grant — minority equity/mezzanine — and should be engaged once an emerging-market project vehicle exists (WaterConnect provides project-preparation support below investment criteria).[^N-26^] Breakthrough Energy carries an explicit caveat: the 500 Mt CO_2e screen is a genuine fit risk for a water/PFAS value proposition, and proposal effort should be conditioned on a credible biosolids-methane accounting first. LIFT costs nothing but time and feeds the US utility credibility that Section 5.3's reliability strategy requires. Two further channels were assessed and deprioritized rather than padded: Xylem's Partnerships Accelerator is invitation-only and unfunded (value is route-to-market; it requires TRL ≥7 and a paying customer — a 2027 item), and UNICEF/UN procurement is a weak product fit for SCWO-scale hardware.[^N-44^][^N-24^]

6.2 Channels accessible via a host partner

The nineteen host-dependent channels share one mechanic: the money flows to a nonprofit, accredited entity, utility, or government borrower, and HCSP participates as technology supplier, JV partner, or project company. They group into four families:

- **Climate funds via accredited entities.** Green Climate Fund project finance (Simplified Approval Process up to $25M per project) requires an Accredited Entity to submit; HCSP enters as technology partner on, for example, a Small Island Developing State wastewater-and-reuse project. The dedicated GCF MSME pilot RFP is closed — watch for successor windows only.[^N-1^][^N-2^] GEF's Non-Grant Instrument (up to $15M per project; $196M GEF-8 blended-finance envelope) flows through partner agencies such as the World Bank, IFC, ADB, and UNIDO.[^N-6^][^N-7^]
- **Multilateral banks via borrowers and projects.** The World Bank's 2030 Water Resources Group structures blended-finance PPPs in active country programs (Bangladesh's industrial water-reuse alliance, Kenya, South Africa, Colombia) and has facilitated $1.6B in water financing; IFC Water finances private clients at scale; IDB Invest/IDB Lab cover Latin America and the Caribbean (industrial wastewater, islands, rum distilleries); the African Development Bank is a GEF implementing agency with >$400M secured; and the Caribbean Development Bank is the strategic procurement channel for the islands vertical. [^N-10^][^N-13^][^N-16^][^N-18^][^N-19^][^N-20^]
- **Corporate foundations via nonprofit implementers.** The Coca-Cola Foundation/RAIN and the Grundfos (Poul Due Jensen) Foundation fund WASH NGOs — e.g., their joint $1.1M 2025 grant to Global Water Center for East Africa — so the access path is partnering with a Water Mission- or Global Water Center-type implementer where HCSP's distilled-water output is the deliverable.[^N-31^][^N-33^][^N-46^] PepsiCo, Nestlé, Diageo, Suntory, and Danone offer no vendor grant windows; engagement is commercial or via the WRG platform — recorded honestly as commercial channels, not grants.[^N-12^][^N-38^][^N-40^]
- **Research and community hosts.** EREF research grants (≤~$300K; emerging contaminants in leachate is an explicit 2024–25 priority) go to academic principal investigators — HCSP co-proposes an SCWO leachate/PFAS-destruction demonstration as site host and co-funder. Rotary Foundation Global Grants ($30K+ budgets, World Fund match $15–200K, rolling) require host and international clubs; viable for small island/community demonstration sites and visibility, not core funding. UNEP-GEF chemicals projects (e.g., the €8.8M+ textile-sector POPs/PFAS program in Bangladesh, Indonesia, Pakistan, and Viet Nam) admit vendors as technology providers within project countries.[^N-65^][^N-66^][^N-22^]

The implication for business development sequencing: host-partner channels are slow (12–36 months from engagement to funded project) but large; they should be seeded now in parallel with the directly actionable channels, not after them. Priority pairings: GCF/CDB for the islands vertical, 2030 WRG Bangladesh for textile-industrial wastewater, and EREF for leachate — each maps to a vertical where no competitor (Section 5) has any presence.

6.3 Not accessible — ruled out, not padded

Six screened channels should receive no further proposal effort: **charity: water** grants only to vetted local nonprofit implementers and its wells/filters portfolio is a technology mismatch;[^N-29^] **WaterAid** is a fundraiser, not a grantmaker — corporate money flows *to* it;[^N-30^] **IRC (WASH)** is a knowledge NGO with no vendor grant program; **KR Foundation** explicitly excludes for-profit organizations and funds advocacy, not market-ready technology;[^N-50^][^N-51^] **Quadrature Climate Foundation** is invitation-only and funds nonprofits/NGOs;[^N-48^] and the **GCF MSME pilot RFP** is closed.[^N-2^] Ecolab Foundation and Xylem

Watermark likewise restrict to nonprofits ($5–50K and partner-grant models respectively); their corporate parents' water capital flows through WaterEquity and commercial partnerships instead.[^N-41^][^N-43^]

6.4 Sequencing with the existing 80-program register

The expanded set stacks onto, rather than replaces, the existing US federal/state/municipal/private register (SERDP/ESTCP, state PFAS programs, Imagine H2O, PRIME, Elemental, WRF). The recommended sequence runs three tracks in parallel. **Track 1 (now, 0–6 months):** submit to the ADB Technology Innovation Challenge and the next 100+ Accelerator cohort; initiate WaterEquity screening if any emerging-market project vehicle exists; lodge a LIFT Technology Scan entry — cash at stake of roughly $550K–$15M for proposal effort measured in weeks. **Track 2 (seed now, harvest 12–36 months):** open accredited-entity conversations for GCF SAP island projects, join 2030 WRG country platforms where textile-industrial wastewater is live, and co-develop an EREF leachate proposal with a university principal investigator — these build the international reference installations Section 5.3 requires before competitor constraints lift. **Track 3 (conditional):** Breakthrough Energy only if the biosolids methane-avoidance case clears the 500 Mt CO_2e screen; ClimateWorks RFPs as they appear; Xylem Partnerships Accelerator once TRL ≥7 and a paying non-grant customer exist. The logic across all three tracks mirrors the competitive thesis: every selected channel monetizes an attribute — modularity, co-product water and energy, small-site economics — that 374Water, GA, Aquarden, and Revive structurally cannot offer, so grant capital compounds the segmentation advantage rather than merely extending runway.

References

Companion reference file for `HCSP_Competitor_Intelligence.md`. Citation indices are prefixed by source research file: `W` = comp_374water.md, `G` = comp_general_atomics.md, `A` = comp_aquarden.md, `R` = comp_revive.md, `N` = grants_ngo.md. Entries below are copied from the reference lists of those verified research files (September 2026); only sources actually cited in the chapter are listed.

374Water Inc. (AirSCWO) — from research/comp_374water.md

[^W-1^]: Quartz/SEC summary of FY2024 10-K (filed 2025-03-28) — https://qz.com/374water-inc-scwo-reports-earnings-1851773002 (Mar 28, 2025). “revenues of $445,445… net loss of $12,434,114… working capital of $11,760,131 and an accumulated deficit of $28,387,618… raised approximately $11.4 million through a registered direct offering.”

[^W-2^]: 374Water FY2025 Form 10-K via last10k (SEC, filed 2026-03-31) — https://last10k.com/sec-filings/scwo. “revenue totaled $0.2 million… reduction in equipment revenue of approximately $1.9 million [OC San]… Net loss… $21.0 million… Cash… $3.2 million.”

[^W-3^]: 374Water FY2025 Form 10-K, SEC EDGAR (accession 0001654954-26-003094, filed 2026-03-31) — https://www.sec.gov/Archives/edgar/data/933972/000165495426003094/scwo_10k.htm. Going-concern Note 3; ATM sales agreements; Nov 2024 offering terms; Mar 2026 convertible notes; “single significant stockholder” selling; AS system “six (6) and ten (10) tons per day”; AS1 “approximately 1 wet ton of waste per day.”

[^W-4^]: 374Water Investor Presentation 4Q25 (company deck) — https://374water.com/wp-content/uploads/2026/03/374Water-Investor-Presentation-4Q25.pdf. System capacities (AS1 1 t/d @ 0.25–0.5 MGD; AS6 6 t/d @ 0.5–1.5 MGD; AS30 30 t/d @ 4–6 MGD; AS100+ @ 20–40 MGD); WDS site capex $1.5–2.0M/$4.5–6.0M/$7.5–10M; WDS revenue $2M+/$3–5M/$12–20M; “Identified demand totaling nearly $1.8BN”; ATM “~$7.0 million… runway into Q2 2026.”

[^W-5^]: 374Water Q2 2026 results press release (Aug 18, 2026) — https://www.morningstar.com/news/accesswire/1208343msn/. “$2.26 million… approximately $2.0 million [from] the OC San Factory Acceptance Test… begin invoicing the remaining $2.6 million… Cash and cash equivalents: $1.8 million as of June 30, 2026… reduce its operating costs by approximately $3.2 million on an annualized basis.”

[^W-6^]: 374Water Q2 2026 Form 10-Q, SEC EDGAR (accession 0001654954-26-007650, filed 2026-08-14) — https://www.sec.gov/Archives/edgar/data/933972/000165495426007650/scwo_10q.htm. Going concern (“cash balance of $1,776,259… accumulated deficit of $57,204,456”); Olathe PO “$4,880,000 firm fixed price, allocated $3,000,000 to the AirSCWO 6 unit…”; Orlando demo “$812,000”; OC San “costs in excess of billings of approximately $1.9 million”; subsequent events ($60k convertible note July 2026; ex-CFO demand letter).

[^W-7^]: 374Water 424B5 ATM prospectus, SEC EDGAR (filed 2026-08-11) — https://www.sec.gov/Archives/edgar/data/933972/000165495426007452/scwo_424b5.htm. “Up to $50,000,000 Common Stock… Robert’s & Ryan… public float… approximately $27,000,000… approximately $1,142,000 shares of common stock remain available… under our Form S-3.”

[^W-8^]: 374Water Q1 2026 results & highlights (8-K Ex-99.1, May 14, 2026) — https://www.sec.gov/Archives/edgar/data/933972/000165495426004906/scwo_ex991.htm. “purchase order and associated milestone payment of $2.2M from Garney… total contract is valued at more than $4.5M”; Orlando demo “approximately $482,000 in service revenue”; Q1 revenue $551,155, net loss $(4,571,623); USAF subcontract award.

[^W-9^]: DEF 14A special meeting proxy (reverse split), SEC EDGAR (Oct–Dec 2025) — https://www.sec.gov/Archives/edgar/data/933972/000165495425012524/scwo_def14a.htm. “to regain compliance with the Minimum Bid Price Requirement for continued listing on Nasdaq.”

[^W-12^]: 374Water OC San deployment-timeline PR (Feb 13, 2025) — https://finance.yahoo.com/news/374water-announces-timeline-deployment-airscwo-212000017.html. “$1.3 Million in projected revenue…”; AS30 “30 wet tons of biosolids per day”; “up to five AirSCWO 200 systems.” Insider sales table: “YAACOV NAGAR… 26 sales selling 1,322,292 shares for an estimated $1,294,373” (via Nasdaq/Quiver).

[^W-13^]: PowerVerde/374Water merger 8-K (Apr 2021) — https://www.sec.gov/Archives/edgar/data/933972/000157570521000209/pwvi_8k.htm. Founders’ bios; Duke license; Merrell Bros. (MBH) $1,135,000 investment and Kokomo fabrication. Nagar settlement PR (Apr 10, 2025): https://finance.yahoo.com/news/374water-announces-settlement-claim-200100263.html.

[^W-14^]: 374Water company history — https://374water.com/company-history/; Wikipedia SCWO entry. Founded 2018 by Nagar & Deshusses as Duke spin-off; Gates Foundation “Reinvent the Toilet”; Nasdaq 2022; “Global Access” program.

[^W-15^]: Yahoo Finance SCWO key statistics — https://finance.yahoo.com/quote/SCWO/key-statistics/ (accessed 2025-10-09 dataset). Market cap history ($181M Jun 2024 → $45M); insiders 45.29% (pre-split basis), institutions 10.75%.

[^W-16^]: Morningstar SCWO quote (Oct 2, 2026) — https://www.morningstar.com/stocks/xnas/scwo/quote. Close $2.31; 17.72M shares; market cap $41.81M.

[^W-17^]: TickerNerd SCWO statistics (Sept 24, 2026) — https://tickernerd.com/stock/scwo-forecast/. Market cap $54.4M; TTM revenue $1.9M; TTM net income −$20.0M; FCF −$11.3M; 48 employees; 17.7M shares.

[^W-18^]: 374Water DEF 14A (Aug 25, 2026), beneficial ownership table, SEC EDGAR — https://www.sec.gov/Archives/edgar/data/933972/000165495426007870/scwo_def14a.htm. “Yaacov Nagar 3,211,263 18.06%; Marc Deshusses 2,265,245 12.74%.”

[^W-20^]: Maven’s Notebook / Daily Pilot (Feb 2023) — https://mavensnotebook.com/2023/02/06/daily-digest-2-6-… “the district’s board of directors approved the project in 2021 at a cost of $8.7 million.”

[^W-21^]: 374Water Orlando 5-year WDS agreement PR (Mar 10, 2026) — https://www.nasdaq.com/press-release/374water-announces-5-year-agreement-city-orlando-approval-launch-licensed-waste. “the Company did not deliver the [OC San] system in 2025 as previously anticipated”; Orlando license 5-year + two 5-year extensions; >99.95% PFAS destruction.

[^W-22^]: OC San FAT completion PR (Jun 2, 2026) — https://finance.yahoo.com/sectors/energy/articles/374water-advances-solution-worlds-growing-120000117.html. “planned for deployment of the system and team in October of this year.”

[^W-23^]: B2i Digital 374Water profile (Aug 25, 2026) — https://b2idigital.com/374water. Orlando hub 88,000 gal storage, $3–5M/yr at initial scale; Orlando 40 MGD “would require more than thirty times the AirSCWO capacity now installed”; OC San first system “less than one percent”; Olathe 11 MGD “would require ten additional systems.”

[^W-24^]: St. Cloud campaign completion PR (Sep 15, 2026) — https://finance.yahoo.com/technology/articles/374water-completes-600-000-waste-120000104.html; Phase 2 PR (Jul 28, 2026). “$600,000 Waste Destruction Services (‘WDS’) contract with the City of St. Cloud, in partnership with Barr Engineering… supported by… LCCMR and… ENRTF”; “$500,000 and $1.5 million in annual revenue per unit.”

[^W-26^]: 374Water Q2 2025 earnings call coverage (Aug 13, 2025) — https://finance.yahoo.com/news/374water-inc-scwo-q2-2025-071207092.html. NC AFFF contract “valued at about $1 million, with potential for significantly higher value.”

[^W-27^]: DoD ESTCP/DIU Detroit demo completion PR (Oct 21, 2025) — https://finance.yahoo.com/news/374water-completes-commercial-scale-field-123100607.html. “approximately 900 gallons… over 80 hours of runtime” at Clean Earth’s Detroit TSDF; Arcadis prime.

[^W-28^]: Gaya Capital ENV Weekly (Aug 14, 2026) — https://www.gayacapital.com/envweekly/08142026. DIU “Success Memo formally validates 374Water’s AirSCWO… DoD estimates more than 700 military installations require PFAS remediation, with projected cleanup spending exceeding $3 billion.”

[^W-29^]: US Navy NDCEE selection PR (Dec 7, 2022) — https://finance.yahoo.com/news/374water-selected-us-navy-joint-140000148.html. Contract value undisclosed.

[^W-30^]: 374Water standard boilerplate (e.g., GlobeNewswire Feb 13, 2025) — “AirSCWO products destruction capacity scale from 1 ton per day to over 200 tons per day. 374Water offers capital sale and lease options, as well as turn-key destruction services.”

[^W-31^]: (don’t) Waste Water podcast profile of Kobe Nagar — https://dww.show/who/kobe-nagar/. “smallest containerized unit, sized for a community of about 6,000 people… at roughly 30,000-person scale the system becomes a net producer of electricity”; ~$23.8M raised to date.

[^W-32^]: 374Water strategy announcement (Sep 9, 2026) — https://markets.businessinsider.com/news/currencies/374water-unveils-strategy-to-scale-airscwo-into-a-global-waste-destruction-platform-1036533961. “regional infrastructure: centralized facilities capable of accepting biosolids from multiple wastewater treatment plants… 42,000 dry tons… at an estimated disposal cost of $500 to $1000… per dry ton”; Olathe “approximately $4.8 million PO.”

General Atomics EMS (PERSES) — from research/comp_general_atomics.md

[^G-1^]: **Forbes — “Neal Blue & family” profile** (accessed Sep. 2026; page dated 2026-09-15). https://www.forbes.com/profile/neal-blue/ — *“With his brother Linden, he bought the then nuclear energy company from Chevron for a reported $60 million in 1986… The largest private government contractor in the U.S.”*

[^G-3^]: **fed-spend.com — “General Atomics Revenue and Federal Contracts: $41.5B”** (2026-04-13). https://fed-spend.com/blog/general-atomics-federal-contracts-deep-dive — *“Privately held by the Blue family since 1986… It reported $3.2*

billion in revenue in 2024 with approximately 15,000 employees… $41.5 billion in cumulative federal contract obligations, including a single $14.1 billion IDIQ awarded in September 2025… industry analysts estimate the company would be valued at $15-25 billion if it were publicly traded."…

[^G-4^]: **Forbes — General Atomics company overview** (accessed 2026). https://www.forbes.com/companies/general-atomics/ — *"General Atomics Financial Summary… Revenue $3.2B. Sources. FactSet; Forbes."*

[^G-8^]: **Rheinmetall press release** (2026-06-18). https://www.rheinmetall.com/en/media/news-watch/news/2026/06/2026-06-18-rheinmetall-and-general-atomics-explore-co-production-of-vektrex-for-artillery-modernisation — *"GA-EMS is one of four main business segments of General Atomics… specializes in electromagnetic aircraft launch systems (EMALS) and advanced arresting gear (AAG), power generation and energy storage, space systems, nuclear fission energy systems… GA-EMS also provides commercial products and services tar…

[^G-9^]: **GA-EMS presentation (John Follin) to BAE/ARDEC, hosted by CSWAB (PDF)** (2016-12-13). https://cswab.org/wp-content/uploads/2017/03/Supercritical-Water-Oxidation-BAE-Presentation-Dec-2016.pdf — *"22 years and $250M of SCWO development; Built over 15 working SCWO systems; Over 35,000 hours of steady state operations"*; *"3-gpm iSCWO Shipping Configuration 25 feet long by 8 feet wide by 9.5 feet tall. Approximate weight = 20,800 pounds"*; *"Multiple systems can be added in parallel to increase capacity (x 3GPM)"*…

[^G-10^]: **NDIA Future Force Capabilities Conference — John Follin (GA-EMS), "PFAS Waste Destruction using GA's iSCWO" (PDF)** (2025-10-02). https://www.ndia.org/-/media/sites/ndia/meetings-and-events/proceedings/2025/FFC/October%202/Follin_John_%202025%20FFC_2%20October%202025.pdf — *"1990s-2012 Development & Select US Government Uses; 2012-2014 First commercial sale…; Won five large Government PFAS test programs along with continued organic destruction sales. Introduced PERSES name for iSCWO; First two systems are stil…

[^G-11^]: **National Research Council — "Evaluation of Demonstration Test Results of Alternative Technologies for Demilitarization of Assembled Chemical Weapons" (Ch. 3, General Atomics Technology Package)** (2000). https://www.nationalacademies.org/read/9776/chapter/5 — *"Although the SCWO system successfully destroyed organic compounds in the liquids, the results did not demonstrate that the system is capable of operating without frequent shutdowns for repair or cleaning… corrosion and salt plugging both raised concerns…

[^G-13^]: **GA press release — "General Atomics to Demonstrate PFAS Destruction System at DIU-Selected Clean Earth Facility in North Carolina"** (2025-01-07). https://www.ga.com/ga-to-demonstrate-pfas-destruction-system-at-diu-selected-clean-earth-facility-in-north-carolina — *"The Defense Innovation Unit (DIU) has selected the Clean Earth waste facility in Charlotte, North Carolina… Under a previously awarded contract from the DoD Environmental Security Technology Certification Program (ESTCP) through the DIU… The Clean…

[^G-14^]: **GA-EMS standard boilerplate (e.g., Morningstar/ACCESS Newswire)** (2024-12-02). https://www.morningstar.com/news/accesswire/949217msn/general-atomics-awarded-navy-contract-to-advance-long-range-maneuvering-projectile — *"GA-EMS offers an expanding portfolio of capabilities for defense, government, and national security customers. GA-EMS also provides commercial products and services targeting the hazardous waste remediation, oil and gas, and nuclear energy industries."*

[^G-15^]: **GA press release — "General Atomics Demonstrates PFAS Waste Destruction for Invited Federal, State and Waste Management Representatives"** (2023-09-12). https://www.ga.com/iscwo-pfas-destruction-demo-day-success — *"To date, our iSCWO systems have destroyed over 6 million gallons of hazardous and non-hazardous waste with a greater than 99.99% destruction efficiency."*

[^G-16^]: **GA press release — "General Atomics Introduces PERSES Waste Destruction System"** (2025-03-10). https://www.ga.com/ga-introduces-perses-waste-destruction-system — *"PERSES™, the new name for the company's commercial waste destruction system formerly known as the iSCWO… PERSES offers greater throughput capacity… Unlike many lab and prototype systems offered today, GA-EMS has had commercial systems in the field since 2012 and has successfully tested the system to destroy more than 200 waste streams."* Two 2025 s…

[^G-17^]: **GA press release — "General Atomics' iSCWO Process Tested, Verified by EPA For the Destruction of PFAS"** (2023-04-24). https://www.ga.com/ga-iscwo-process-tested-verified-by-epa-for-the-destruction-of-pfas — *"The Environmental Protection Agency (EPA) issued a detailed report (EPA/600/R-22/257) documenting test results and verifying GA-EMS' commercial iSCWO system's effectiveness in the destruction of PFAS… greater than 99.99%… We have been advancing SCWO technology since the 1990s, supplying systems to suppo…

[^G-19^]: **U.S. EPA — "Alternative Treatment Technologies to Open Burning and Open Detonation of Energetic Hazardous Wastes" (final report, Dec. 2019)**. https://www.epa.gov/sites/production/files/2019-12/documents/final_obod_alttechreport_for_publication_dec2019_508_v2.pdf — *"Alkaline hydrolysis… has been in production at Tooele Army Depot, Utah, for many years… The iSCWO neutralization technology was added to treat secondary waste streams resulting from the base hydrolysis process but eventually was not used due to la…

[^G-20^]: **Wikipedia — Blue Grass Chemical Agent-Destruction Pilot Plant** (accessed Sep. 2026). https://en.wikipedia.org/wiki/Blue_Grass_Chemical_Agent-Destruction_Pilot_Plant — *"In August 2020, the decision was made not to use the supercritical water oxidation system to process hydrolysate… In March 2021, program officials announced that hydrolysate… would be destroyed at Veolia North America near Port Arthur, Texas."*

[^G-21^]: **National Research Council — "Analysis of Engineering Design Studies for Demilitarization of Assembled Chemical Weapons at Blue Grass Army Depot" (Ch. 5)** (2002). https://nap.nationalacademies.org/read/10509/chapter/7 — BGCAPP design with four SCWO reactors and brine-recovery system; off-site disposal of SCWO effluent under investigation.

[^G-22^]: **NATO/CCMS SCWO fact sheet (PDF)** (2002-12-11). https://www.ihpa.info/docs/library/reports/nato/NATO_SCWOFactSh&Annex10.pdf — *"A commercial scale 2 gpm General Atomics demonstration system is operating in Japan under a joint development agreement with… Kurita Water Industries, Ltd and Komatsu, Ltd… operated more than 1000 hours and at commercial scale at 1 ton/hour (8 t/day)… Pilot tests up to 227 kg/hr. Full-scale Blue Grass design is 2700 kg/hr."*

[^G-23^]: **UNEP/Basel Convention — "General technical guidelines on the environmentally sound management of POPs wastes" (UNEP/CHW.15/6/Add.1; also COP17 revision)** (English PDF, accessed 2026). https://www.basel.int/Portals/4/download.aspx?d=UNEP-CHW.15-6-Add.1-Rev.1.English.pdf — *"General Atomics Electromagnetic Systems (GA-EMS) have developed an industrial SCWO system (iSCWO) which [is] transportable and housed in shipping containers… Transportable iSCWO is currently used in Kyrgyzstan and Tajikistan to destroy the…

[^G-24^]: **GA press release — "General Atomics Awarded Contract for Hazardous Waste Destruction Systems in China"** (2018-04-17). https://www.ga.com/general-atomics-awarded-contract-for-hazardous-waste-destruction-systems-in-china — *"Awarded a contract from Tianjin Saipeng Environmental Protection Technology Company… for three iSCWO systems… installed at the Binhai New Area facility."*

[^G-25^]: **GA press release — "Defense Innovation Unit Awards General Atomics a Contract to Demonstrate iSCWO System for PFAS Destruction"** (2023-12-11). https://www.ga.com/ga-awarded-defense-innovation-unit-contract-to-demonstrate-iscwo-system-for-pfas-destruction — *"GA-EMS will deliver and install a commercial-scale iSCWO system at a DoD-identified site to conduct on-site PFAS destruction testing starting in early 2024."*

[^G-27^]: **GA press release — "General Atomics PERSES Waste Destruction System Scheduled to Destroy PFAS at Lake Elmo, Minnesota Site"** (2025-04-16). https://www.ga.com/ga-perses-waste-destruction-system-scheduled-to-destroy-pfas-at-lake-elmo-minnesota-site — *"GA-EMS was awarded a contract by Bay West LLC… leading the research program funded by the U.S. Army Engineer Research and Development Center (ERDC)… The system will be at the Lake Elmo site for about 4 weeks and then removed to other sites for additional tests un…

[^G-28^]: **Waste Advantage Magazine — "General Atomics Demonstrates PFAS Destruction System for Minnesota Congresswoman, State and Local Officials"** (2025-06-10). https://wasteadvantagemag.com/general-atomics-demonstrates-pfas-destruction-system-for-minnesota-congresswoman-state-and-local-officials/ — confirms Bay West/ERDC program and on-site Lake Elmo demos.

[^G-31^]: **EnviroWiki — "Supercritical Water Oxidation (SCWO)"** (page updated 2026-03-02). https://www.enviro.wiki/index.php?title=Supercritical_Water_Oxidation_(SCWO) — *"Early projections for treatment costs (CAPEX + OPEX) for slurries are in the range of $12 to $90 per ton (or $0.04 to $0.37 per gallon) depending on system scale and contaminant concentration, with a majority of the cost coming from amortizing the equipment."* (Generic SCWO economics, not GA-specific pricing.)

[^G-32^]: **IPQwery — General Atomics trademark portfolio** (accessed 2026-01). https://www.ipqwery.com/ipowner/en/owner/ip/510712-general-atomics.html — PERSES USPTO serial 99221721, filed 2025-06-06, **registered 2026-01-13**, classes 7/11/37/40 (waste destruction machines, wastewater treatment apparatus, installation/repair, treatment of materials).

[^G-33^]: **Wikipedia — Supercritical water oxidation (commercial applications)** (accessed Sep. 2026). https://en.wikipedia.org/wiki/Supercritical_water_oxidation — *"Aquarden is also providing remediation of hazardous energetic wastes and chemical warfare agents with SCWO, where a full-scale SCWO system has been operating for some years in France for the Defense Industry."* Also Chematur/Johnson Matthey UK AquaCat unit (3,000 L/h, "the largest SCWO unit in the world"). No GA–ArianeGroup connection documented.

Aquarden Technologies (SuperOx) — from research/comp_aquarden.md

[^A-1^]: Proff.dk — AQUARDEN TECHNOLOGIES ApS, Regnskab (annual reports 2021–2025, CVR/Virk data). URL: https://www.proff.dk/regnskab/aquarden-technologies-aps/skc3a6vinge/producenter/GQE5V1I016D (accessed Sep 2026). Verbatim (DKK '000): "Årets resultat: -11.177 / -10.886 / -10.029 / -4.343 / -13.025"; "Egenkapital i alt: -21.505 … -30.947"; "Antal ansatte: 13 / 13 / 8 / 10 / 11"; "Likvide midler … 2.706"; "Anden gæld … 76.769".

[^A-2^]: cvrapi.dk — AQUARDEN TECHNOLOGIES ApS. URL: https://cvrapi.dk/virksomhed/dk/aquarden-technologies-aps/29151911 (accessed Sep 2026). Verbatim: “CVR-nummer 29151911. Startdato 31/10 - 2005 … Ejere, Cai Holding … Direktører, Zhuoyan Cai”.

[^A-3^]: Proff.dk — Roller, AQUARDEN TECHNOLOGIES ApS. URL: https://www.proff.dk/roller/aquarden-technologies-aps/skc3a6vinge/producenter/GQE5V1I016D (accessed Sep 2026). “Direktør: Zhuoyan Cai; Stifter: Zhuoyan Cai; Revisor: Dansk Revision Hillerød.”

[^A-5^]: Aquarden — History & timeline. URL: https://aquarden.com/about/timeline/ (2024). “2024: Second industrial scale SuperOx SCWO plant operational in France with ArianeGroup for PFAS destruction … 2018: First industrial scale SCWO plant installed in France with our partner ArianeGroup … 2005: Aquarden Technologies founded in Denmark.”

[^A-6^]: Aquarden — SuperOx® SCWO. URL: https://aquarden.com/superox-scwo/ (2024). “Throughput 5 tonnes/day per unit … Operates solely on electricity. No fossil co-fuels required … Modular design.”

[^A-7^]: IFAT exhibitor presentation — “SuperOx® – SCWO Aquarden Technologies”. URL: https://exhibitors.ifat.de/en/exhibitors-products/products-services/products-services-details/productsDetail/2080/ (accessed Sep 2026). Verbatim: “Family-owned, Est. 2005 … Denmark, Singapore … Commercialization via select partners in France and North America – partnership with ArianeGroup”; “Field deployable: onsite or containerized … One reactor: Up to 12 m3/day liquid wastes/slurries … Energy consumption: 80-300 kWh per m3 wastewater”…

[^A-8^]: Aquarden — Aerospace & Defence. URL: https://aquarden.com/applications/aerospace-defence/ (2024). “Aquarden has provided our SuperOx® SCWO system for two of ArianeGroup’s French sites: the first industrial scale SuperOx® unit was operational in 2018 … followed by a second unit in 2024 for PFAS destruction … Our partnership began in 2017 through ArianeGroup’s innovative remediation project ELIXIR™.”

[^A-9^]: ArianeGroup — “ArianeGroup revolutionizes industrial waste management with Elixir™” (9 Feb 2024). URL: https://www.ariane.group/en/news/arianegroup-elixir-revolutionizing-industrial-waste-management/. Verbatim: “in partnership with Danish start-up Aquarden Technologies”; “A facility such as this can treat between 100 and 150 tons of solid waste and around 1,500 m3 of liquid waste”; “contained within a 200 m2 area, and can be dismantled and moved in less than two weeks”; “ArianeGroup’s next Elixir™ facility will…

[^A-10^]: French Ministry for Ecological Transition — Plan eau, 3 ans après (Innov’Eau). URL: https://portail.documentation.developpement-durable.gouv.fr/pub/MPDOUV00268987-plan-eau-ans-apres-100-des-mesures-initiees-dont-m.html (accessed Sep 2026). Verbatim: “Développé par ArianeGroup, ELIXIR est un procédé d’oxydation sous eau supercritique (SCWO) qui détruit > 99,99% des PFAS … Evalué à un coût total de 5,03 M€, ce projet lauréat Innov’Eau a bénéficié d’une aide de 1,59 M€.”

[^A-11^]: Krause et al. (2022), “Supercritical Water Oxidation as an Innovative Technology for PFAS Destruction,” ASCE J. Hazardous, Toxic, and Radioactive Waste / PMC10428202 (US EPA ORD-directed study). URL: https://pmc.ncbi.nlm.nih.gov/articles/PMC10428202/. Verbatim: “PFOS was reduced from 26.2 mg/L to 240 µg/L … from the Aquarden … demonstration”; “For the demonstration, Aquarden used a feed flow of 1 L/h and a reactor residence time of 60 s.”

[^A-12^]: State of Green — “SCWO technology eliminates 99.9% of PFAS from wastewater” (9 Mar 2020). URL: https://stateofgreen.com/en/news/scwo-technology-eliminates-999-of-pfas-from-wastewater/. “Arlanda in Sweden has identified PFAS as a pollutant in their soil and water, and has therefore engaged Aquarden Technologies to remove these compounds.”

[^A-13^]: State of Green — “PFAS Destruction Through Supercritical Water Oxidation” (11 Nov 2020). URL: https://stateofgreen.com/en/news/pfas-destruction-through-supercritical-water-oxidation/. “A project with Stockholm Arlanda Airport similarly exhibited SCWO’s destruction efficiency with PFAS from a concentrated feed containing 679 µg/L PFOS and 34 µg/L PFOA, where both substances were eliminated to below their detection limits of 0.01 µg/L.”

[^A-15^]: Danish Environmental Protection Agency (MUDP) — “PFAS-inator: Sustainable and cost-effective technology for PFAS-removal from landfill leachate,” final report (Dec 2024; revised Jun 2026, ISBN 978-87-7038-692-0). URL: https://groenprojektbank.dk/Media/639183181296586761/Slutrapport%20-%20PFAS-inator.pdf. Verbatim: “under the Danish Environmental Agency’s grant ‘MUDP 2020’ … executed between February 2020 and December 2023”; “Aquarden’s fullscale SuperOx® plant can typically handle continuous treatment flowrates…

[^A-16^]: VIO.nu — PFAS-håndbogen (Danish PFAS handbook, 2022). URL: https://www.vio.nu/wp-content/uploads/2023/03/PFAS-haandbogen_2022-1.pdf. Verbatim (transl.): SCWO “is used inter alia in the Korsør case … also being trialled for leachate treatment at Danish landfills. The method is expensive and is therefore used as the final step in a treatment train…”

[^A-17^]: CleantechWatch — “Underskudsramt cleantech-selskab vil skabe forretning på PFAS-forurenet spildevand” (27 Sep 2021). URL: https://ctwatch.dk/nyheder/milj_teknik/article13305844.ece.

[^A-18^]: KFF Health News — “Inside the Pentagon’s Painfully Slow Effort to Clean Up Decades of PFAS Contamination” (21 Dec 2023). URL: https://kffhealthnews.org/public-health/pentagon-defense-department-pfas-contamination-cleanup-slow/. Verbatim:

"said Zhuoyan Cai, director of Denmark-based Aquarden Technologies, which he said is currently in talks with U.S. companies about its SCWO technology."

[^A-19^]: nano tech 2026 seminar — PFAS Busters LLC / Dr. Hiroshi Tomimasu (29 Jan 2026). URL: https://unifiedsearch.jcdbizmatch.jp/nanotech2026/en/sem/nanotech_mems/seminar_details/jnjY4V17wjU. "founded PFAS Busters LLC to introduce Aquarden Technologies' Supercritical Water Oxidation (SCWO) technology from Denmark to Japan"; "The SuperOx system has been deployed in more than five countries in Europe."

[^A-20^]: REGENESIS / Environment Analyst — "Corporate Guide: Identifying and Remediating PFAS" (2025). URL: https://regenesis.com/wp-content/uploads/2025/08/Environment-Analyst-Corporate-Guide-Identifying-and-Remediating-PFAS-2025.pdf. "SCWO units, such as those developed by General Atomics, Aquarden and 374Water, are at commercial launch stages. But they require high capital investment and skilled operation, often limiting their early adoption to larger regional facilities or remediation contractors."

[^A-21^]: Sci-Bot — Global PFAS industry market assessment (26 Jun 2026). URL: https://sci-bot.ru/conduct-comprehensive-market-assessment-of-578a. "high capital cost (~USD 2–10M per unit)"; "USD 100–500/ton of concentrate"; "Aquarden demonstrated >99% PFAS removal for Swedish landfill leachate and Stockholm Arlanda Airport AFFF concentrate."

[^A-27^]: ArianeGroup — ELIXIR™ datasheet (Apr 2026). URL: https://www.ariane.group/app/uploads/2026/06/datasheet-elixir-en-version2026_prov-basseresolution.pdf. "END-TO-END SERVICE … FIXED OR MOBILE FACILITIES, DIRECT PURCHASE OR LEASING."

[^A-29^]: Aquarden — brochure download page (2026). URL: https://aquarden.com/download/. "The Aquarden SuperOx® enables fire services, airports, military sites, and industrial facilities to permanently remove PFAS e.g. from legacy firefighting foam stocks…"

Revive Environmental (PFAS Annihilator) — from research/comp_revive.md

[^R-1^]: Business Wire / Battelle — "Battelle, Viking Global Investors Launch Revive Environmental," Jan 17, 2023. https://www.battelle.org/insights/newsroom/press-release-details/battelle-viking-global-investors-launch-revive-environmental (also https://revive-environmental.com/battelle-viking-global-investors-launch-revive-environmental-2023/)

[^R-2^]: Signal Net — "Quiet Breakout: The Strongest Bond in Organic Chemistry Has Six New Ways to Break," Jun 5, 2026. https://www.signalnet.ai/quiet-breakout-the-strongest-bond-in-organic-chemistry-has-six-new-ways-to-break/

[^R-5^]: BlueTech Research Innovation Tracker — "Revive Environmental" (snapshot: Service Based/Treatment Fee; Private; Seed; Early Revenue). https://www.bluetechresearch.com/tools/innovation-tracker/revive-environmental/

[^R-6^]: GlobeNewswire — "Revive Environmental Announces CEO Transition," Mar 6, 2026. https://finance.yahoo.com/news/revive-environmental-announces-ceo-transition-220800443.html

[^R-8^]: Revive Environmental blog citing Associated Press — "PFAS is Everywhere—And the World is Paying Attention" (AP interview, Mar 2025). https://revive-environmental.com/pfas-is-everywhere-and-the-world-is-paying-attention/

[^R-9^]: Democracy in NH / NH Bulletin — NH-Revive contract $668,258 for 10,000 gal AFFF; Heritage-Crystal Clean hauls to Wyoming, MI. Aug 2023. https://democracynh.com/?page=32 ; https://newhampshirebulletin.com/2023/07/25/nh-is-first-state-to-partner-with-pfas-annihilator-to-destroy-firefighting-foams/

[^R-10^]: Office of Ohio Governor — "Governor DeWine Announces Expansion of Firefighting Foam Takeback Program" (Round 1: $3M Monsanto PCB settlement; ≥60,000 gal remaining; registration to Sept 4, 2026). https://governor.ohio.gov/wps/portal/gov/governor/media/news-and-media/governor-dewine-announces-expansion-of-firefighting-foam-takeback-program

[^R-11^]: Gaya Capital ENV Weekly, May 1, 2026 — "New Jersey / Revive Environmental | $16.6m program appropriation; contract value not disclosed | ~150,000 gallons of AFFF from more than 400 departments." https://www.gayacapital.com/envweekly/05022026

[^R-13^]: GlobeNewswire — "Revive Environmental First to Complete Verified PFAS Destruction Under North Carolina's AFFF Take-Back Program," Jan 22, 2026. https://www.globenewswire.com/news-release/2026/01/22/3224234/0/en/

[^R-14^]: GlobeNewswire — "EPA Administrator Lee Zeldin Tours Revive Environmental's Columbus Headquarters, the Nation's First Commercial PFAS Destruction Facility," Jul 7, 2026 (5 Annihilator units; "cost-effective at scale"). https://revive-environmental.com/epa-administrator-lee-zeldin-tours-revive-environmentals-columbus-headquarters-the-nations-first-commercial-pfas-destruction-facility/

[^R-16^]: Business Wire — "Revive Environmental PFAS Annihilator Deployed in First-to-Market Commercial Destruction of 'Forever Chemicals'," May 1, 2023 (Grand Rapids; 300–500 gpd concentrate; 160k gpd raw leachate). https://www.businesswire.com/news/home/20230501005063/en/

[^R-17^]: Battelle — “West Michigan facility operating first commercial-scale PFAS-destroying technology,” May 5, 2023 (Wyoming, MI; 4never; >100k gpd leachate). https://www.battelle.org/insights/newsroom/news-details/2023/05/05/west-michigan-facility-operating-first-commercial-scale-pfas-destroying-technology

[^R-18^]: Revive Environmental — “Revive Environmental’s Columbus Facility Now Fully Permitted to Treat PFAS” (second permitted US site; first in Ohio), 2026. https://revive-environmental.com/revive-environmentals-columbus-facility-now-fully-permitted-to-treat-pfas/

[^R-19^]: NPR — “A new tool is helping to annihilate dangerous PFAS in firefighting foam,” Mar 16, 2026 (Ohio collected ~14,000 gal spring 2024; Ohio EPA seeking funds for more). https://www.npr.org/2026/03/16/nx-s1-5714017/

[^R-20^]: Business Wire — “Revive Environmental Wins 2025 WEFTEC Innovative Technology Award for PFAS Annihilator SCWO,” Sep 16, 2025. https://www.businesswire.com/news/home/20250916378962/en/

[^R-22^]: Revive Environmental — “Ohio AFFF Takeback Program Complete,” Aug 11, 2026 (13,500+ gal; 120 fire departments/airports; $3M Monsanto settlement). https://revive-environmental.com/ohio-collects-and-destroys-large-quantities-of-firefighting-foam-after-takeback-program-throughout-the-state/

[^R-23^]: Carolina Connection (UNC) — “A state pilot program to destroy PFAS has entered its final phase,” Apr 23, 2025 (4 vendors: AxNano, Clean Earth, Revive, 374Water; 374Water projects $100–$400/gal). https://carolinaconnection.org/2025/04/23/a-state-pilot-program-to-destroy-pfas-has-entered-its-final-phase

[^R-24^]: Gaya Capital ENV Weekly, May 1, 2026 — “Colorado / Aquagga | Contract not disclosed; $40/gal program payment | … using HALT.” https://www.gayacapital.com/envweekly/05022026

[^R-25^]: Business Wire — “Revive Environmental and Battelle Partner on DoD-Funded Project to Eliminate Toxic ‘Forever Chemicals’” (ESTCP/DIU demo at Clean Earth, Detroit), Apr 2, 2025. https://www.businesswire.com/news/home/20250402919534/en/

[^R-26^]: Mitchell Williams Law — analysis of DoW Feb 26, 2026 report: >2M gal AFFF + 1.5M gal rinsate; AFFF use prohibition extended to Oct 1, 2026. https://www.mitchellwilliamslaw.com/detailed-analysis-of-options-for-the-destruction/disposal-of-covered-materials-containing-per-and-polyfluoroalkyl-substances-us-department-of-war-report

[^R-27^]: Waste Dive — “DOD guidance recommends hazardous waste landfills, incineration to manage PFAS,” Mar 17, 2026 (SCWO listed among “emerging options”). https://www.wastedive.com/news/dod-guidance-hazardous-waste-landfills-incineration-manage-pfa/814937/

[^R-28^]: Nasdaq/Heritage-Crystal Clean — “Heritage-Crystal Clean, Allonnia, Revive Environmental, and EPOC Enviro Launch 4never™,” Apr 5, 2024 (hub-and-spoke; fixed cost per gallon). https://www.nasdaq.com/press-release/heritage-crystal-clean-allonnia-revive-environmental-and-epoc-enviro-launch-4nevertm

[^R-29^]: Revive Environmental — “Revive Environmental’s Impact at the 2025 ARFF Symposium” (airport AFFF/rinsate projects; Certificate of Annihilation, EPA Method 1633). https://revive-environmental.com/revive-environmentals-impact-at-the-2025-arff-symposium/

[^R-31^]: GlobeNewswire — “Revive Environmental Ready to Support DoW PFAS Destruction Requirements Under New Federal Guidance,” Mar 11, 2026 (“state agencies, DoW installations, Fortune 500 industrials, and municipal utilities”; “nation’s only commercially permitted PFAS destruction facilities”). https://www.globenewswire.com/news-release/2026/03/11/3253844/0/en/

[^R-34^]: Revive Environmental — “Landfill Leachate Treatment and PFAS Destruction” (on-site concentration; concentrate shipped to permitted facility; plug-and-play deployment “within weeks”). https://revive-environmental.com/landfill-leachate-pfas-services/

[^R-35^]: MLive — “Forever no more: New tech ‘annihilates’ toxic PFAS chemicals,” Aug 15, 2023 (shipping-container units; 15–40 ¢/gal; 6 units in production, up to 25; Crystal Clean 11 candidate facilities). https://www.mlive.com/public-interest/2023/07/forever-no-more-new-tech-annihilates-toxic-pfas-chemicals.html

[^R-36^]: ES&T Water (peer-reviewed), 2025 — spent-media economics; centralized reactivation contracts limited to large facilities (>36,000 kg/changeout); Scenario A off-site recovery cost-effective for “small- to medium-scale or fragmented systems.” https://pmc.ncbi.nlm.nih.gov/articles/PMC12260480/

[^R-37^]: Wisconsin Examiner — “System to ‘annihilate’ PFAS chemicals deployed in Michigan,” Jun 27, 2023 (~500 gpd concentrate; ~160–165k gpd raw leachate). https://wisconsinexaminer.com/2023/06/27/system-to-annihilate-pfas-chemicals-deployed-in-michigan/

[^R-38^]: Revive Environmental via Water Journal (Korea) — EPA interim disposal guidance PR, Apr 2024 (“capacity to destroy up to 7 million gallons of PFAS laden waste annually… price competitive with incineration”). https://www.waterjournal.co.kr/news/articleView.html?idxno=74941

[^R-39^]: Ohio OBM contract document (Battelle/Revive AFFF takeback scope) — collection events limited to <1,000 gal per waste owner; larger volumes require separate arrangements. https://api.obm.ohio.gov/files/download/44/eb81621a-9451-4df6-a3cf-652064421a2b/inline

[^R-40^]: Waste Dive — “DOD-funded PFAS demo projects show promise for remediation and destruction,” Dec 1, 2025 (on-site more efficient/cost-effective than off-site due to logistics/permitting; Revive treated AFFF in Detroit). https://www.wastedive.com/news/dod-pfas-destruction-disposal-demos-waste/805991/

[^R-43^]: Minnesota PCA — “Groundbreaking study shows unaffordable costs of PFAS cleanup from wastewater” (small facilities’ per-pound costs >6× large facilities). https://www.pca.state.mn.us/news-and-stories/groundbreaking-study-shows-unaffordable-costs-of-pfas-cleanup-from-wastewater

[^R-44^]: Route Fifty — “Small water utilities cannot achieve PFAS cleanup on their own,” Sep 30, 2024 (95% of systems <10,000 customers; ≤3 FTE staff). https://www.route-fifty.com/infrastructure/2024/09/small-water-utilities-cannot-achieve-pfas-cleanup-their-own/399605/

Expanded non-dilutive funding channels — from research/grants_ngo.md

[^N-1^]: UNFCCC/GCF — “Accessing the Green Climate Fund” (SAP up to USD 25M; funding windows) — https://unfccc.int/sites/default/files/resource/3.%20%5BGreen%20Climate%20Fund%5D_NBF_Session%205_GCF_20241129.pdf — 2024-11-29

[^N-2^]: Green Climate Fund — Other funding options (REDD+ and MSME pilot RFPs ended) — https://www.greenclimate.fund/access-funding/other-funding-options — accessed Sept 2026

[^N-4^]: GEF — Challenge Program for Adaptation Innovation (direct engagement with private-sector innovators; USD 20M across two initial calls; winners implemented with GEF agencies) — https://www.thegef.org/what-we-do/topics/challenge-program-adaptation-innovation — accessed Sept 2026

[^N-5^]: GEF press release — “$20 million investment in climate adaptation innovators” (>USD 40M across 32 projects; +USD 40M to 2026; LDCF/SCCF) — https://www.thegef.org/newsroom/press-releases/gef-announces-20-million-investment-climate-adaptation-innovators — 2024-11-15

[^N-6^]: FundsforNGOs — 4th Call, GEF-7 Non-Grant Instrument Program (max USD 15M/project; debt/equity/guarantees; Partner Agency submission) — https://www2.fundsforngos.org/latest-funds-for-ngos/4th-call-for-proposals-gef-7-non-grant-instrument-program/ — 2021-01-18

[^N-7^]: GEF — Report of the GEF to the UNFCCC COP (GEF-8: USD 196M to blended finance; biannual calls; SCCF-B tech transfer/private sector) — https://unfccc.int/ttclear/misc_/StaticFiles/gnwoerk_static/tn_poznan/077c785fb367458780568d9770f4098e/d358d92f81574ac4bfd894c6642f3cbd.
— 2025

[^N-10^]: World Bank — 2030 WRG 2025 Annual Report (Gazipur wastewater PPP: USD 40M public + ~USD 45M private; A4R reuse alliance; USD 370M supported 2023–25; USD 1.6B mobilized) — https://www.worldbank.org/en/topic/water/publication/2030-water-resources-group-wrg-2025-annual-report — 2026-05-26

[^N-12^]: UNDP India / 2030 WRG — joint press release (global partners incl. Nestlé, PepsiCo, Coca-Cola, Dow, AB InBev; AfDB, IDB) — https://www.undp.org/india/press-release-joint-publication-2030-wrg-world-bank-undp-unveils-innovative-and-sustainable-partnership-and-financing-models-towards-gender-and — accessed Sept 2026

[^N-13^]: IFC — Water sector page (>USD 5B committed/mobilized since 1994; debt, equity, advisory, concessional finance) — https://www.ifc.org/en/what-we-do/sector-expertise/infrastructure/water — accessed Sept 2026

[^N-15^]: ADB — Technology Innovation Challenge (grant up to USD 450,000 per pilot; 10% co-financing; test/validate in new markets) — https://challenges.adb.org/en/challenges/tic — accessed Sept 2026

[^N-16^]: IDB Invest / PR Newswire — inaugural CHF 100M Blue Bond for sustainable water management in LAC — https://www.prnewswire.com/news-releases/idb-invest-launches-inaugural-swiss-franc-bond-diversifying-investor-base-and-expanding-privatesector-financing-in-latin-america-and-the-caribbean-302786190.html — 2026-06-01

[^N-18^]: IDB — “IDB and IDB Invest provided $26 billion in climate financing over five years” (incl. IDB Lab innovation instruments) — https://www.iadb.org/en/news/idb-and-idb-invest-provided-26-billion-climate-financing-over-five-years — 2022-11-10

[^N-19^]: African Development Bank — GEF partnership page (AfDB as GEF Implementing Agency; >USD 400M secured for member countries) — https://www.afdb.org/en/topics-and-sectors/initiatives-partnerships/global-environment-facility-gef — 2026-06-05

[^N-20^]: OECD — Caribbean Development Dynamics 2025 (debt-for-climate-resilience swap w/ EIB & IDB guarantees tied to sovereign water-security project) — https://www.oecd.org/content/dam/oecd/en/publications/reports/2024/12/caribbean-development-dynamics-2024_86de96a9/a8e79405-en.pdf — 2024-12

[^N-22^]: Bureau Veritas webinar — UNEP-GEF project (EUR 8.8M+) on POPs/PFAS reduction in textiles (Bangladesh, Indonesia, Pakistan, Viet Nam); innovation ecosystem — https://bureau-veritas.easywebinar.live/registration-443 — accessed Sept 2026

[^N-24^]: Josh's Water Jobs — UNICEF RFP/2025/01/9198242 climate-resilient WASH, Kerala (UNGM registration; e-submission) — https://www.joshswaterjobs.com/jobs/330731/ — 2025-07-02

[^N-26^]: Access Newswire — "WaterEquity Announces $100M in Financing Available for Water and Sanitation Infrastructure" (open application; tickets $2–15M equity/mezz; wastewater treatment/reuse and C&I projects; WaterConnect project prep) — https://www.accessnewswire.com/newsroom/en/banking-and-financial-services/funding-for-growth-waterequity-announces-100m-in-financing-available-942405 — 2024-11-14

[^N-27^]: Xylem/WaterEquity — Water & Climate Resilience Fund >$100M raised of $200M target (Microsoft anchor; Starbucks, Xylem strategic; Ecolab, Reckitt, Gap) — https://www.xylem.com/el-gr/about-xylem/newsroom/press-releases/waterequity-announces-$100m-raised-for-new-impact-fund/ — accessed Sept 2026

[^N-29^]: Nonprofit Pro — charity: water grantmaking (>$80M/yr in grants; funds local partner organizations only) — https://www.nonprofitpro.com/post/grant-writing-101-charity-water-shares-3-strategies-it-uses-to-effectively-fund-its-programs/ — 2022-01-12

[^N-30^]: Josh's Water Jobs — WaterAid Strategic Partnership Development Manager (secures corporate donations ≥£100K/yr — funding flows to WaterAid, not from it) — https://www.joshswaterjobs.com/jobs/379141/ — 2026-06-25

[^N-31^]: GreenEconomy.Media — Coca-Cola RAIN South Africa ($1.25M grants to five catchment projects, 2019; replenishment focus) — https://greeneconomy.media/coca-colas-rain-set-to-replenish-15-billion-litres-of-water-over-the-next-decade/ — accessed Sept 2026

[^N-33^]: Coca-Cola — Global Water Center East Africa project: US$1.1M in grants from Grundfos Foundation and The Coca-Cola Foundation — https://www.coca-cola.com/ke/en/media-center/east-africa-water-project-launch — 2025-03-20

[^N-34^]: Procurement Magazine — AB InBev 100+ Accelerator (up to USD 100K per startup pilot; water stewardship challenges; 116+ companies) — https://procurementmag.com/articles/what-is-ab-inbevs-100-accelerator-programme — 2024-01-31

[^N-35^]: Startup Grants India — 100+ Accelerator profile (cross-corporate: AB InBev, Coca-Cola, Colgate-Palmolive, Danone, Mondelēz, Unilever; 240+ companies, 40+ countries) — https://www.startupgrantsindia.com/providers/100-accelerator — accessed 2026-07-10 snapshot

[^N-38^]: Food Digital — Diageo water security (Spirit of Progress; up to £5M/5yr Scotland peatland restoration; replenishment) — https://fooddigital.com/news/diageo-promote-water-security-drinks-sector — 2026-02-20

[^N-40^]: CEO Water Mandate Water Action Hub — Suntory Natural Water Sanctuaries ("Mizu To Ikiru"; forest groundwater recharge at plant catchments since 2003) — https://wateractionhub.org/projects/1128/d/suntory-natural-water-sanctuary-okudaisen-and-kasara/ — accessed Sept 2026

[^N-41^]: Ecolab — 2025 U.S. Regional NPO Grant Program guidelines (501(c)(3) required; four focus areas incl. environment/water) — https://www.ecolab.com/-/media/Ecolab/Ecolab-Home/Documents/DocumentLibrary/CommunityRelations/NPO-Grants/Regions/2025/Ecolab-Apply-for-Grant-Information-Regions-for-web-posting-APPLICATION-OPEN-2025-pdf.pdf — 2025-05-29

[^N-43^]: Xylem — Watermark page (nonprofit partners: UNICEF USA, Planet Water, Mercy Corps, EWB; community grants to 148 nonprofits in 2023) — https://www.xylem.com/it-it/watermark/ and https://xyleminc.gcs-web.com/news-releases/news-release-details/xylem-teams-partners-and-communities-globally-accelerate — accessed Sept 2026

[^N-44^]: Xylem — Partnerships Accelerator (TRL ≥7, ≥1 paying customer, invitation-only, applications ~October; 12-month Go/No-Go commercial pathway) — https://www.xylem.com/en-bg/about/xylem-innovation-labs/partnerships-accelerator/ — accessed Sept 2026

[^N-46^]: PMC/World Pumps — Grundfos Foundation overview (DKK 200M donations 2020; Water/Research/Inclusion; NGO partners Water Mission, UNICEF, Oxfam, Plan) — https://pmc.ncbi.nlm.nih.gov/articles/PMC7180044/ — accessed Sept 2026

[^N-48^]: UNDP BIOFIN FIRE database — Quadrature Climate Foundation (eligible: nonprofits/NGOs; nomination process) — https://fire.biofin.org/single/quadrature-climate-foundation — updated 2025-01-09

[^N-50^]: KR Foundation — How we work (no unsolicited applications; invitation-only) — https://krfnd.org/how-we-work/ — accessed Sept 2026

[^N-51^]: Fund for Bangladesh — KR Foundation grant criteria ("does not support for-profit organizations"; DKK 2–5M grants) — https://fundforbangladesh.wordpress.com/2018/09/15/ — 2018-09-15

[^N-52^]: Hinchilla — ClimateWorks Foundation (no open application; periodic targeted RFPs; eligibility has included private enterprises) — https://www.hinchilla.com/funders-us/262303250-climateworks-foundation — accessed Sept 2026

[^N-53^]: AndPurpose — ClimateWorks Adaptation and Resilience Fund RFP (USD 9M pool; deadline Sept 19, 2025) — https://andpurpose.world/climateworks-foundation-adaptation-and-resilience-fund-for-communities-facing-climate-risks/ — 2025-08-25

[^N-54^]: OpenGrants — Deep-tech non-dilutive funding guide (Breakthrough Energy Fellows Cohort 7: Innovator up to $500K via SAFE; Explorer $50–250K grants TRL 0–2; 500Mt CO2e screen; next deadline Nov 2, 2026) — https://opengrants.io/guides/deep-tech-non-dilutive-funding/ — 2026-09-23

[^N-55^]: Inside Philanthropy — Breakthrough Energy Discovery (Fellows, Workshops, Ecosystem Grants; philanthropic funding for early-stage climate tech) — https://www.insidephilanthropy.com/find-a-grant/grants-b/breakthrough-energy — 2026-07-01

[^N-60^]: The Daily Pennsylvanian — Water Center at Penn launches Water-AI Nexus Center of Excellence with AWS, WEF, Leading Utilities of the World (Climate Week NYC) — https://www.thedp.com/article/2025/10/penn-amazon-water-ai-center-climate-change-ai — 2025-10-03

[^N-61^]: WEF/WERF — LIFT program booklet (Technology Evaluation Program; utility-industry vetting system) — https://www.wef.org/globalassets/assets-wef/2-resources/topics/a-n/innovation/technical-resources/ipbooklet15.pdf — accessed Sept 2026

[^N-64^]: NEBRA/WEF — Call for innovative wastewater technologies, LIFT Technology Scan (resource-recovery intensification forum) — https://www.nebiosolids.org/resource-recovery-forum-2015 — accessed Sept 2026

[^N-65^]: Funds for Individuals/NGOs — EREF Research Grants RFA (priorities incl. emerging contaminants in leachate disposal/treatment; typical ≤$300K, ~2 yrs; Dec 1, 2024 deadline cycle) — https://fundsforindividuals.fundsforngos.org/grants/request-for-applications-eref-research-grants/ — 2024-11-18

[^N-66^]: Rotary Club of Port of Spain — Rotary grants overview (global grant minimum budget $30,000; World Fund $15,000–$200,000; DDF matched 100%, cash 50%) — https://rotaryclubpos.org/Stories/what-you-should-know-about-rotary-grants — accessed Sept 2026